

**PROJECT TITLE: PREDICTIVE ANALYTICS FOR DIGITAL
MARKETING CAMPAIGN SUCCESS**

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Declaration

I hereby declare that this dissertation titled “**Predictive Analytics for Digital Marketing Campaign Success**” is my original work submitted in partial fulfilment of the requirements for the **MSc Data Science, AI and Digital Business** programme.

I declare that this research represents my own work and that all sources of information, ideas and reference quoted in this dissertation are acknowledged. I state that this thesis has not been presented for examination or any other academic qualification at any university or other relevant institution before.

The content, analysis, findings, and conclusions of this dissertation are my own and do not necessarily represent the views of the organization(s) nor the U.S. Government.

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Abstract

This research explores predictive analytics and machine learning solutions that can be used to predict the success of digital marketing campaigns. The study generates and tests predictive models which leverage secondary marketing data for the purpose of pinpointing key performance data, optimizing campaigns, and facilitating informed data-driven marketing decisions.

Literature review regarding predictive analytics, machine learning (ML) and prediction of the success of digital marketing campaigns was provided in this chapter. It reviews the challenges faced during the performance evaluation process, predictive strategies, machine learning models, evaluation methods, theoretical perspectives and research gaps related to developing data-driven marketing solutions.

This study follows a quantitative research process and its dataset employed in this research is secondary Digital Marketing dataset which consists of customer demographic data, purchasing behaviour, customer interaction with campaign, attributes related to customer response. Data preprocessing, exploratory data analysis, feature engineering, developing machine learning models, and evaluating the models. Three machine learning algorithms are crafted: Logistic Regression, Decision Tree and Random Forest are all designed and compared, to gain an insight of how effective they are in predicting the outcome of the campaign.

The results show that predictive analytics can provide insightful data points for campaign success, including previous interactions, spending data, previous purchases and interaction data with customers. The comparative analysis identifies the observed differences in the performance of the two models and how machine learning techniques could help to forecast campaigns with increased accuracy. In addition, the feature importance analysis gives hints about the variables that prominently affect the prediction results.

The research in this paper adds to the study of digital marketing analytics with an integrated solution that incorporates predictive modelling, performance assessment, and marketing interpretation. The research also underscores the significance of data quality, selecting the correct model, and ongoing assessment when applying machine learning solutions in today's fast-changing digital marketing landscape.

Keywords: Predictive Analytics, Digital Marketing, Machine Learning, Campaign Success, Data-Driven Decision Making, Feature Engineering, Marketing Optimisation.

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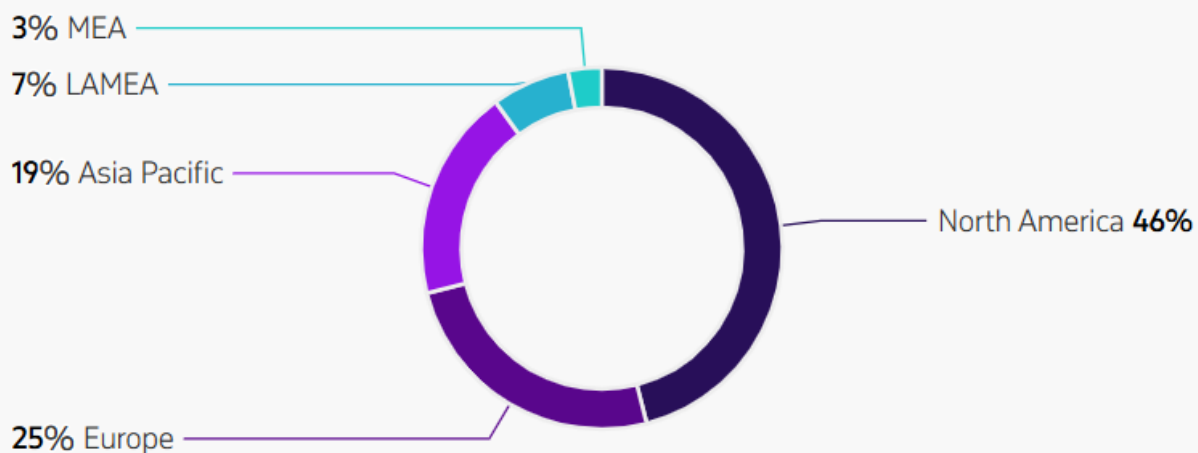
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Chapter 1: Introduction

1.1 Background of the Study

Digital marketing businesses seek to engage with customers, market effectively, and achieve online success, digital marketing has become an essential element of business strategies. In the era of digital, where the organizations are keen to reach their customers and promote their products and products are an integral part of the business strategy. Looking forward to 2030, the global predictive analytics market is expected to grow at a compound annual rate (CAGR) of 16.4% from 2024 to 2030 reaching \$82.35 billion by 2030. The growth reflects more and more companies are taking advantage of predictive analytics for better business forecasting, decision and streamlining. (Itransition, 2026). Digital channels have exponentially grown and given rise to enormous amounts of data to inform marketers about their ad's campaigns, such as how much they have invested, how many times their ads get viewed, how much customer engagement data they gather and how long their campaigns run. With this data comes the possibilities of employing analytical methods to determine the effectiveness of their campaigns and how to make more informed marketing decisions.



Scheme title: Predictive analytics adoption across regions

Data source: Precedence Research

Figure 1.1: Predictive analytics adoption across regions

(Source: Itransition, 2026)

Adoption of predictive analytics is mostly high in North America (46%), followed by Europe (25) and Asia Pacific (19) (Itransition, 2026). The data reveals that currently, developed regions are more advanced in adoption, with low- and medium-developed countries slowly following the trend of using predictive analytics to boost business growth in these emerging regions. Marketing predictive analytics solutions enable huge amounts of data to be processed, with the aim of forecasting the outcome of promotions (Gülter & Cevher, 2025). Organisations can leverage predictive analytics to gain insight into data and fine-tune their campaigns, adjust customer experience, and use data-based ideas to enhance the performance of their marketing.

Success in a Digital Marketing campaign means when your marketing actions bring about the desired results: higher engagement, more conversions and high ROI on your advertising dollars. Predictive Analytics adoption is growing, with an uptick of 30% in customer engagement tools, to enable better proactive engagements with customers (Itransition, 2026). Businesses are seen using predictive models to improve customer retention, engagement and communication between them and customers. Organisations are often struggling to effectively monitor and forecast the performance of the campaign caused by evolving customer behaviour, competitive market and poor marketing resource allocation (Berlilana *et al.*, 2025). The traditional ways of analysing performance are usually limited to analysing past campaign performance, and this does not allow in-depth predictions about future campaign performance.

Predictive analytics goes beyond this with more sophisticated tools like statistical techniques, data mining and machine learning algorithms that use past marketing data to sift through and “predict” the success of future campaigns. Organisations can discover key performance metrics, fine-tune ads, refine their target audience and aid in data-centric decision-making with predictive analytics (Sangsawang, 2024). This way, marketers can transition from analysing data to planning campaigns.

The predictive marketing ability has also been enhanced by machine learning applications, which let models learn from prior marketing campaign details on their own and supply accurate forecasting. Marketing factors such as Logistic Regression, Decision Tree, and Random Forest can

analyse the relationships between the marketing factors and outputs of a campaign to find important variables that then influence the campaign's success. Utilising machine learning-driven predictive analytics offers valuable opportunities to boost the effectiveness of digital marketing, minimise the use of resources and optimise strategic decision-making (Ikeh, 2025). One of the focus areas of this research is to understand predictive analytics and machine learning strategies to predict the performance of digital marketing campaigns and the important factors that contribute to the performance.

1.2 Problem statement

The amount of information people generates when working with digital marketing is enormous, but it's difficult to keep track of the results of their campaigns and make accurate forecasts about a campaign's success. Limiting factors in making decisions includes identifying the factors that can influence your organization, optimize marketing resources and choose the best predictive method that can help.

1.3 Research Aim

The aim is to investigate predictive analytics and machine learning techniques to yield accurate and meaningful predictions on the success of digital marketing campaigns.

1.4 Research Objectives

- To identify the important factors of digital marketing that determine how successful a campaign is and to how well digital marketing is working overall.
- To perform data quality and model performance analytical pre-processing tasks with secondary marketing data and touch up the data as necessary.
- To develop and create Machine Learning Models of digital marketing campaigns based on previous data patterns for accuracy of prediction.
- To evaluate models' performance (accuracy and classification), against actual data using suitable evaluation metrics.

1.5 Research Questions

RQ1: How can predictive analytics techniques be applied effectively to forecast digital marketing campaign success outcomes?

RQ2: Which digital marketing performance factors significantly influence campaign success and overall marketing effectiveness?

RQ3: Which machine learning model achieves the highest prediction accuracy for digital marketing campaign success?

RQ4: How can predictive analytics improve marketing decision-making through data-driven campaign optimisation strategies?

1.6 Research Rationale

As the use of digital marketing platforms rises, tremendous amounts of data have been created from the campaigns. But it can be hard for organisations to come to an accurate conclusion about the results of a campaign and their accuracy at predicting future outcomes based on this data (Sadriana, 2023). This research is carried out in order to fill the gaps in the need for predictive approaches that can identify valuable marketing performance factors, which can assist in campaign decision-making. The study will use predictive analytics and machine learning techniques to gain insight into factors that contribute to the success of a campaign, and create models capable of predicting success. This research that is important in the following aspects: Improving the efficiency of marketing, allocation of resources and strategic planning based on data.

1.6 Research Significance

The study proves to be of academic value and is of great practical importance as it investigates the utilisation of predictive analytics and machine learning in the optimisation of digital marketing campaigns. The study offers useful insight into how well machine learning models can utilise marketing data and forecast key parameters that affect marketing campaign effectiveness, especially for academic research. The results can guide organisations to make informed decisions based on the evidence, optimise ad spend, and improve the business perspective of campaign planning. Decision tree reviews comparing the predictive models also provide some indication of the success rating of each machine learning marketing prediction

technique (El-Hajj & Pavlova, 2024). So, the significant to the marketing analytics body and an assistance tool for implementing a data-driven business strategy.

1.7 Research Scope

The purpose of this study is to explore on how the performance of digital marketing campaigns can be analysed through predictive analysis and Machine Learning techniques. This study is based on secondary marketing data, which contains elements of the marketing campaign such as advertising investment dollars, number of impressions, rates of clicks, conversion rates and campaign outcomes. The research included the following parts, from data preprocessing to the evaluation of the model performance with suitable measures. The machine learning methods that are studied here are Logistic Regression, Decision Tree and Random Forest models (Omar *et al.*, 2024). Traditional, historical data is used for predicting a campaign's success, with no consideration of marketing optimisation, primary surveys of customers or any live campaigns being conducted.

1.8 Dissertation Structure

Chapter 1: Introduction

Chapter 2: Literature Review

Chapter 3: Research Methodology

Chapter 4: Data Analysis and Findings

Chapter 5: Conclusion and Recommendations

Figure 1.2: Dissertation Structure

Chapter 2: Literature Review

2.1 Introduction

This chapter is the detailed literature review of previous work on predictive analytics and machine learning techniques used for prediction of the success of digital marketing campaigns. It covers some of the important research topics such as factors affecting the success of digital marketing, methods for prediction analytics, machine learning models, and methods used to evaluate digital campaigns to predict outcomes. Critically evaluates past studies and finds out knowledge developments, limitations and research gaps. Moreover, the chapter puts forth the theories adopted for the research that are relevant for technology uptake and data-informed decision making. The results obtained from the literature review are used to guide the formulation of research methodology and approach in analysing the problem.

2.2 Digital Marketing Performance Factors Influencing Campaign Success and Effectiveness

The rise in dependency on the Internet for organisations to capture clients, elevate engagement and meet business goals has rendered digital marketing an important business process. According to Dutta, (2024), The development of digital platforms has allowed businesses to gather a considerable amount of information regarding campaigns, such as how much they spend on advertising, when they engage customers, impressions, click-through rates, conversion rates and engagement behaviour. But in recent years, customer behaviours and the competitive digital world have grown more complex, making it difficult for organisations to determine what factors are the most influential when it comes to the campaigns' success. The key performance indicators that digital marketing involves (Massoudi *et al.*, 2023). Thus, the factors affecting the performance of digital marketing have become a significant study to better plan and make decisions about digital marketing campaigns.

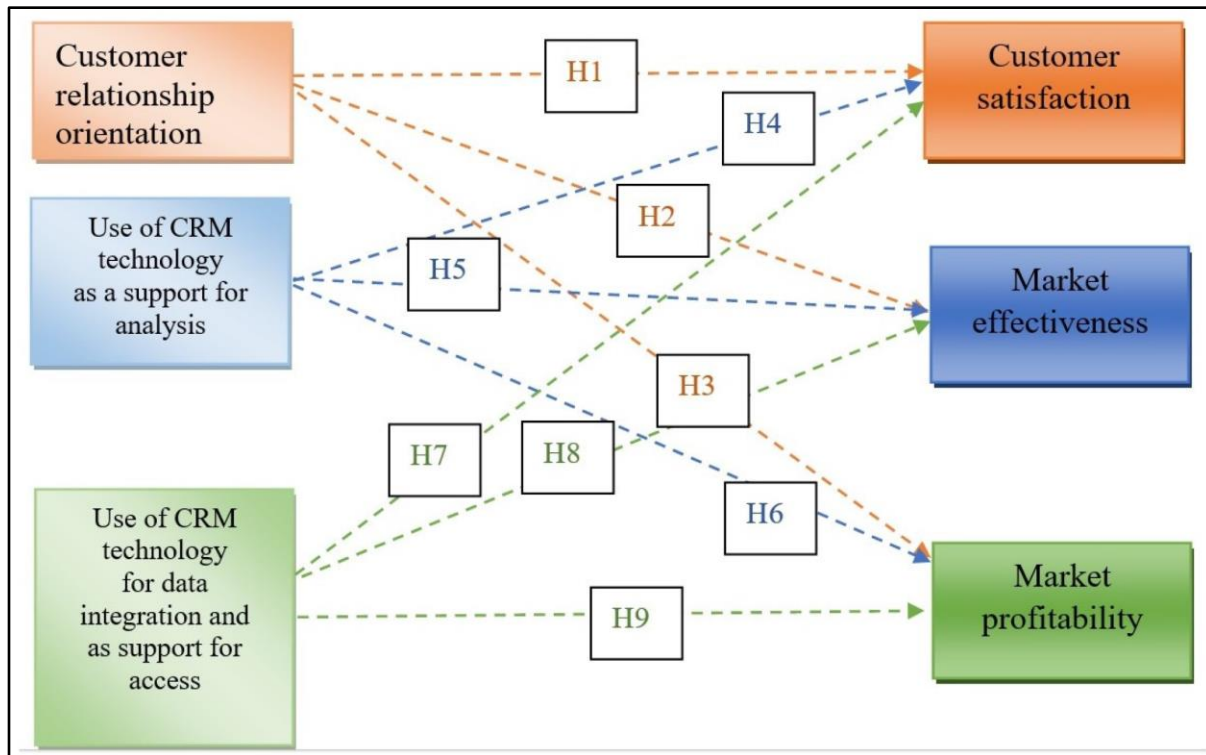


Figure 2.1: The Effects of a Digital Marketing Orientation

(Source: Paşcalău *et al.*, 2024)

Typically campaign success is measured by multiple measures of active performance, not one measure. According to Rahardja and Aini, (2025), Customer Engagement, Customer conversion, ROI and audience response are some of the common metrics of Marketing effectiveness. Metrics like: clicks, likes, shares, and interactions show the interest of your customers and give you information about the involvement of your audience when marketing content. Likewise, conversion-related measures such as conversions, bounce rate, click-through rate and observed behavior help to determine if campaigns prove effective at persuading customers to take the action(s) desired. While these will be useful metrics, looking at each of them on their own may not give the full picture (Berlilana *et al.*, 2025). Multiple factors need to be considered to establish connectivity between marketing activities and generation of a positive outcome.

Advertising is listed among the key determinants of campaign performance and is a factor if budgets are allocated that will impact a campaign's reach and visibility. Targeting the right audience and designing a campaign and selecting the right platforms and preferences of the

customers are all important factors in the effectiveness of budget utilisation (Cynthia *et al.*, 2025). Groups must develop methods that enable them to identify if there is any benefit in boosting investment or if more might be better spent elsewhere. Thus, a combination of analysis of advertising expenditure with other campaign attributes represents a fuller understanding of the effectiveness of the marketing. Success of a digital marketing campaign is also driven by factors that relate to the audience. How audiences react to ads is influenced by their behaviours, demographics, preferences and internet interactions. Consumers' expectations, however, have changed and their behaviours in the online world are constantly evolving so that alone is not enough to rely on traditional segmentation approaches (Gülter and Cevher, 2025). By using data-driven analysis, organizations can uncover patterns that exist behind customer interactions and use the insights to target new ones.

A campaign's length and the time of year are other considerations that could have an impact on marketing results. Customer exposure, engagement and performance are all factors that are influenced by the length of a campaign (Hayadi and El Emary, 2024). Being able to build an immediate response may come from a short campaign, or a longer campaign can lead to higher brand awareness and customer relationships. Likewise, the effectiveness of the campaigns also can vary by platforms and markets. By knowing more about these factors, an organisation can effectively devise a better marketing strategy and optimise how the campaigns are performed.

Overall, existing literature shows that financial, behavioural and operational factors can trigger the success of the digital marketing campaign. Once the performance indicators are identified it becomes easier to appreciate the importance of these indicators and more possible to use machine learning techniques for more accurate predictions or forecasts with relevant features (Almestarihi *et al.*, 2024). Hence, the analysis of these factors is crucial in creating effective predictive analysis solutions and thus facilitating better campaign optimisation and decisions based on data.

2.3 Predictive Analytics Applications for Digital Marketing Campaign Forecasting and Optimisation

Organizations need to make better decisions in making use of digital marketing, and this is where predictive analytics has become an essential method for making predictions based on historical and real-time data. While traditional, 'analytical' methods primarily focus on 'what has happened,'

“predictive” analytics uses mathematical models, statistical analysis, data mining and machine learning techniques to detect patterns and predict future trends. According to Marković, (2026), a growth of customer interaction data from digital platforms, which provides opportunities for organisations to gain a clearer understanding of the customer's behaviour, predict the effectiveness of campaigns and fine-tune marketing efforts. With increasing competitiveness in digital marketing, predictive analytics empowers companies to make proactive decisions instead of just making observations in the past.

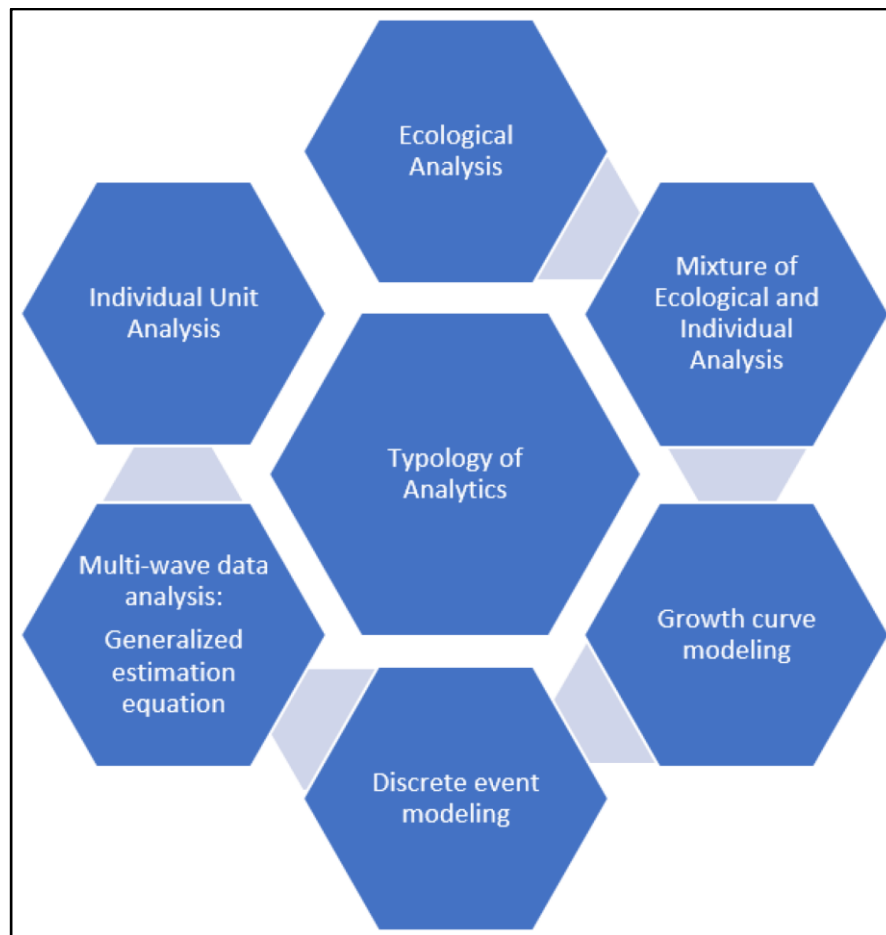


Figure 2.2: Predictive Analytics with a Transdisciplinary Framework

(Source: Wan and Wan, 2023)

In digital marketing, predictive analytics uses multiple data sources about a marketing cause to determine a relationship among marketing activities and anticipated results. Osakwe *et al.*, (2023),

Organizations can leverage predictive models for predicting customer responses, customer conversion rates, target high-value customer groups and optimise advertising campaigns, amongst other use cases. Previous campaign characteristics can be analysed to ascertain what factors have been associated with success and advice can be recommended for future campaigns based on that (Al Rafi, 2023). A feature of this may help marketers to target more effectively, lower waste on marketing dollars and boost marketing effectiveness.

Customer behaviour forecasting is among the major use cases for predictive analytics. According to Ikeh, (2025), digital platforms gather significant behavioural data such as viewing history, engagement interactions, purchase history and preference for interaction. These patterns can be analysed by predictive models, which can give a grasp of the customers' intentions and anticipation of future actions. For instance, organisations can anticipate consumers that are likely to interact with certain ads or make purchases (Al Khaldy *et al.*, 2023). This can help businesses deliver relevant content to targeted audiences, which helps to inform and execute personalised marketing approaches. But the key for achieving the successful predictive steps are the accurate data collection, proper features selection and suitable analytical tools.

Predictive analytics also plays a role in optimizing the campaigns, enabling organizations to assess the potential outcomes of their marketing decisions. Predictive models can help businesses make educated guesses about possible effects when they adjust their advertising spend, advertising-to-consumer targeting, length of the ad campaign, and communications style (Sadrnia, 2023). While it offers immense advantages to digital marketing practices, there are several challenges to implementing predictive analytics. A data quality issue is also of concern as data completeness/inconsistencies and inaccuracy may affect the credibility of predictions (Chattopadhyaya, 2026). Additionally, customers' purchasing patterns can be complicated due to the fast-changing market trends, social factors and technological advancements. Organisations need to regularly update their analytical models, and select variables that employ customers' current behaviour (Theodorakopoulos and Theodoropoulou, 2024). Also, predictive models can be complex, which might pose challenges for marketing team members who are not technically proficient in data analytics.

Overall, the literature suggests that predictive analytics offers substantial value in digital marketing, and it can help turn collected data into actionable insights. It can help a company predict

campaign results, target customers more accurately, optimize resource use, and aid in empirical decision-making processes. The predictive methods would need to be analyzed in detail to evaluate their performance in predicting the success of a digital marketing campaign (Ghodake *et al.*, 2025). Thus, the study of predictive analytics' applications is a critical stepping stone towards the design of solutions that will boost marketing effectiveness through the use of machine learning.

2.4 Machine Learning Models for Predicting Digital Marketing Campaign Outcomes

Machine learning (ML) is a widely adopted technique in predictive analytics that has earned significance due to its capability to handle massive amounts of data, recognize intricate patterns and extract precise forecasts from historical data. According to Ikeh, (2025), in the world of digital marketing, machine learning models are being employed more and more to forecast campaign results based on the correlation between some marketing metrics and performance measures. In contrast, traditional statistical methods are based on predetermined assumptions, while machine learning algorithms can learn from the data and enhance their prediction capacity based on pattern recognition (Berlilana *et al.*, 2025). It has provided huge potential for organisations to make smarter use of data-driven approaches to campaign planning, customer targeting, marketing optimisation, etc.

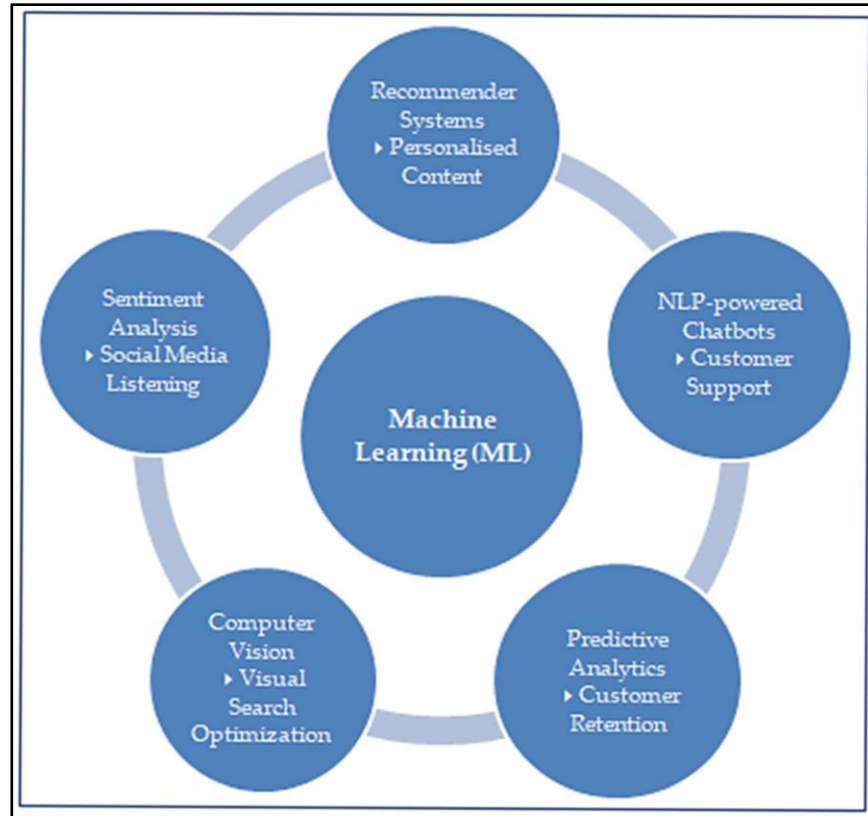


Figure 2.3: Machine learning applications in Digital Marketing

(Source: Kaponis *et al.*, 2025)

One of the most notable benefits of machine learning in digital marketing is its capability to analyze a number of campaign-oriented factors all at once. Typical ad metrics on digital marketing datasets are ad spend, impressions, customer engagement, click through rates, conversions and campaign length (Güler and Cevher, 2025). Analysis of interactions between these variables can help identify the factors that most influence the success of the campaign, using machine learning models. Logistic Regression is a widely used machine learning model for prediction, given a labelled dataset with categories. Logistic Regression can predict successful or unsuccessful outcomes in digital marketing relevant applications, using the factors which determined the outcome of past campaigns (Hayadi and El Emary, 2024). Interpretability is desirable for the model, as it reveals the impact that each variable has on the predictions. This is especially well suited to marketing decision makers who need specific questions answered about how, and why,

certain aspects of a campaign perform or don't. But, Logistic Regression has limitations in highly complex relationships as it normally considers linear relationships between variables.

An alternative technique is Decision Tree algorithms, where decisions are displayed in a framework that is structured like a tree. Decision Trees can be helpful to discover significant decision rules and to see how the various combinations of marketing factors affect success (Cynthia *et al.*, 2025). For instance, it can determine if an increased level of advertising spending, engagement rate or conversion is correlated with achieving the goal for a successful ad campaign. Random Forest is a challenging ensemble learning technique which further enhances the predicting power of a model which is formed by stacking multiple Decision Trees together. Random Forest builds trees than the single predictive tree and outputs the average value of many trees to get a more consistent and reliable prediction (Sangsawang, 2024). This way, the risk of overfitting is minimized, and those models will have better generalisation skills when confronted with unseen data. Random Forest can give relative importance of campaign features, and come up with improved predictive performance in digital marketing prediction tasks over individual models.

Machine learning has taken a more significant place in marketing analytics, especially in recent years with the developments in artificial intelligence. Advanced models have the capability to process more extensive and intricate data, which enables companies to create personalised marketing strategies and boost customer engagement (Prasetio *et al.*, 2025). Most of the previous works, however, are mostly devoted on predicting customer responses or predicting clicks on advertisements rather than predicting the actual success of the whole marketing campaign by employing multiple machine learning models. The strengths and weaknesses of the Logistic Regression, Decision Tree and Random Forest models are quite different in their interpretability, flexibility and predictive power. By exploring these models, organisations can gain deeper insights into the mechanisms by which they can leverage machine learning techniques to enhance campaign decision-making and ensure effective digital marketing strategies.

2.5 Comparative Evaluation of Predictive Models Using Performance Measurement Metrics

Predictive modelling and comparison of predictive models are crucial steps to improve the effectiveness of machine learning based solutions in digital marketing campaigns and make

predictions. According to Gupta, *et al.* (2024), while machine learning algorithms offer many potent tools for analysing marketing data, choosing the right one demands a systematic approach, leveraging the right performance measures. Depending on the characteristics of the data, relationships between the features, and the way the algorithm is designed, various algorithms may yield different predictions

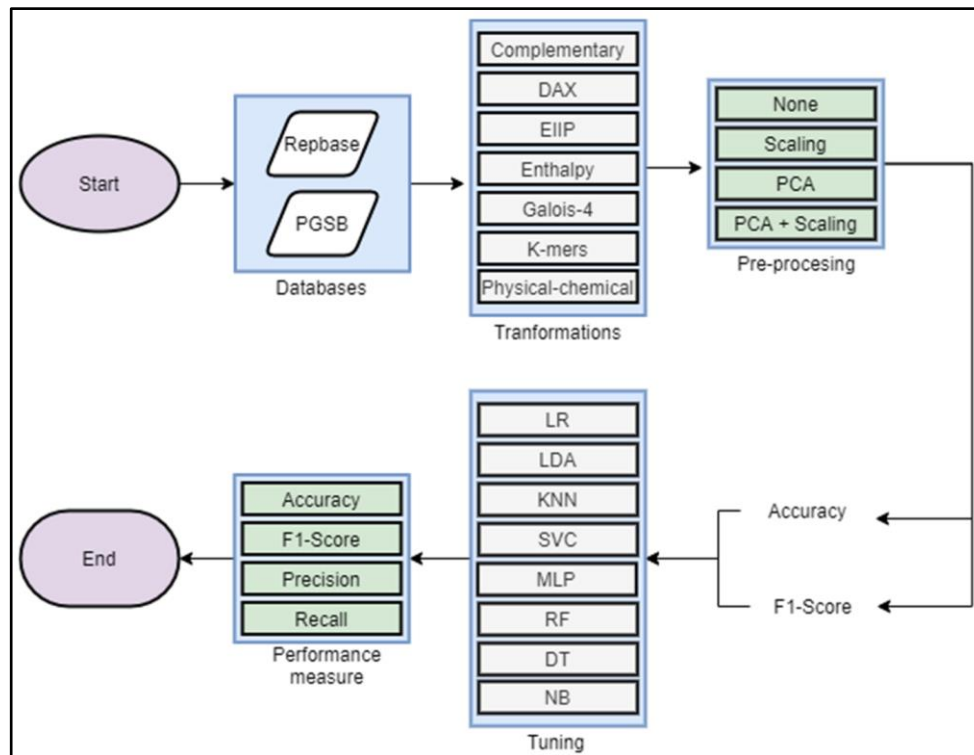


Figure 2.4: Performance Metrics of Machine Learning Algorithms

(Source: Influenced by Mundu *et al.*, 2026)

In predictive analytics, model evaluation typically involves assessing how accurate models are in categorizing or predicting results. According to Mundu *et al.* (2026), two of the more commonly used assessment metrics are accuracy and recall. Accuracy is the ratio of the number of instances correctly predicted divided by the total number of instances observed. Recall is another commonly-used criterion, representing the number of predicted instances correctly divided by all the instances observed (Tousi and Luján, 2022). Also, accuracy can be misleading as there might be an unequal distribution in the test data between the various outcomes of campaigns.

There are other measurements of the predictive model effectiveness: precision and recall. Precision is the ratio of correctly predicted successful campaigns, divided by the total of all campaigns predicted successfully (Rahman and Haque, 2022). The significance of the metric is that predictions may be incorrect, resulting in organisations investing in campaigns that are unlikely to get the result they need. Recall is the extent to which a model can recognise all the true successful campaigns. A model with a high amount of recall helps the organisation see that a potentially effective marketing opportunity isn't being lost (Malik, and Rana, 2025). Hence, considering precision and recall is better to come to a conclusion about the predictive capability.

The F1 score is a performance measure that is a balanced combination of precision and recall. This metric can be of particular help if organizations must take false positive/false negative predictions into account. Errors of prediction in digital marketing campaign analysis can lead to ineffective spending on advertising, misallocation of resources, or missed opportunities (Modak *et al.*, 2024). Thus, the F1-score is useful as an indicator of the overall reliability of predictive models in addition to representing accuracy. Another crucial evaluation method is a confusion matrix analysis, as it can give in-depth details regarding prediction results (Bellar *et al.*, 2024). It shows the number of True Positive, True Negative, False Positive, False Negative cases. This helps researchers to be aware of the advantages and disadvantages of predictive models. For instance, the model can be very accurate for the overall quality of the model, but have a large number of predictions of campaigns being successful that are wrong (Ziakakis, and Vlachopoulou, 2023). These shortcomings can be noted through the use of the confusion matrix, which can assist in understanding and in making a more educated decision regarding which model to choose.

Model comparison is also used in understanding the significance of the factors involved in marketing performance. Input variables obtained from feature analysis techniques can identify the most important features for successful and successful campaign predictions. This information has obvious practical applications as it allows organisations to target their resources on the factors that can have the most impact on the outcomes. Thus, predictive model performance evaluation not only determines which algorithm is working the best, but it also provides marketing strategy insights for optimisation.

The assessment is an important genre for digital marketing analytics research, and it is essential that comparative assessment be carried out between two or more predictive models. The accuracy,

precision, recall, F1_score and analysis of the confusion matrix give a detailed view of the performance of the model (Soltaninejad, *et al.*, 2024). This study aims to determine the best machine learning technique to develop the model that can predict the success of digital marketing campaigns and propose data-informed marketing strategies through comparing various machine learning techniques.

2.6 Theoretical framework

2.6.1 Technology Acceptance Model (TAM)

Organisations usually will use predictive analytics and machine learning in digital marketing decision-making, the technology acceptance model (TAM) can offer a suitable theoretical foundation. The theory states that two key factors in the acceptance and use of new technologies are perceived usefulness and perceived ease of use (Mukred *et al.*, 2024). For digital marketing predictive analytics, perceived usefulness is how organisations evaluate the ability of the predictive models to enhance the effectiveness of campaign forecasting and the utilisation of marketing resources, as well as the effectiveness of informed decision making. If marketers understand that the machine learning model could make accurate predictions and solve their problems, they will more likely use them in their marketing.

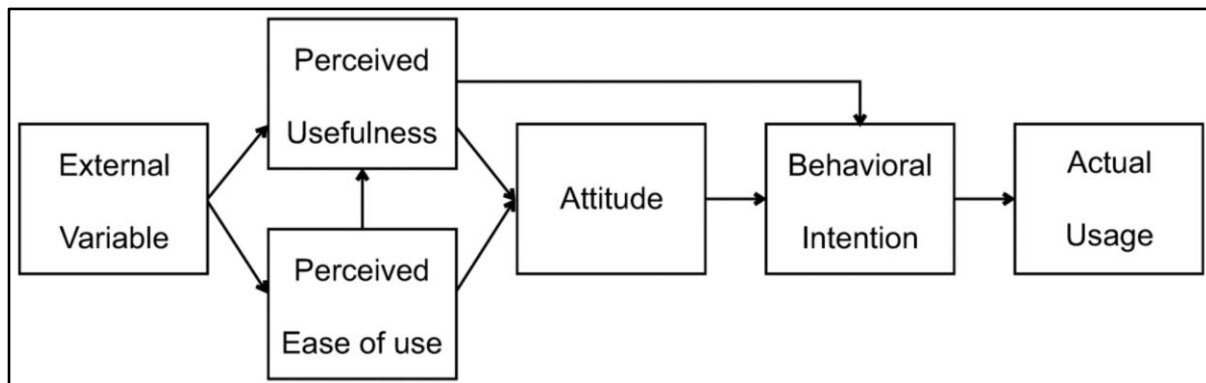


Figure 2.5: Factors of the Technology Acceptance Model

(Source: Influenced by Mukred *et al.*, 2024)

Perceived ease of use is defined as the ease of system implementation and use, as it is perceived by the user. Predictive analytics solutions have more chances of being adopted by digital marketing

professionals when models create helpful outputs and actionable suggestions (Phuong Dung *et al.*, 2023). However, TAM applies to this research; as the successful application of predictive analytics heavily relies not only on the performance of a technical model, there is also an organizational acceptance and implementation process. The theory helps to give explanations about how the predictive modelling tools can affect and influence the marketing strategies (Song, *et al.*, 2025). The confidence to enhance the data-driven decision making process, and prompt organizations to adopt the advanced analytical methods instead of traditional analysis methods.

2.6.2 Data-Driven Decision-Making Theory

Another important theory that was used in this research is the Data-Driven Decision-Making Theory because it outlines the usage of data analysis methods to assist and enhance organisation strategic decisions and operation. The theory further argues that decisions using reliable data and analytical evidence are more effective in making decisions than decisions based only on intuition and experience (Elgendy *et al.*, 2022). For digital marketing, companies amass considerable amounts of information on the campaigns they run, including ads spend, impressions, rates of engagement, clicks, and conversions. This data can be harnessed to provide actionable insights by way of predictive analytics, which can assist in forecasting and optimising campaigns.

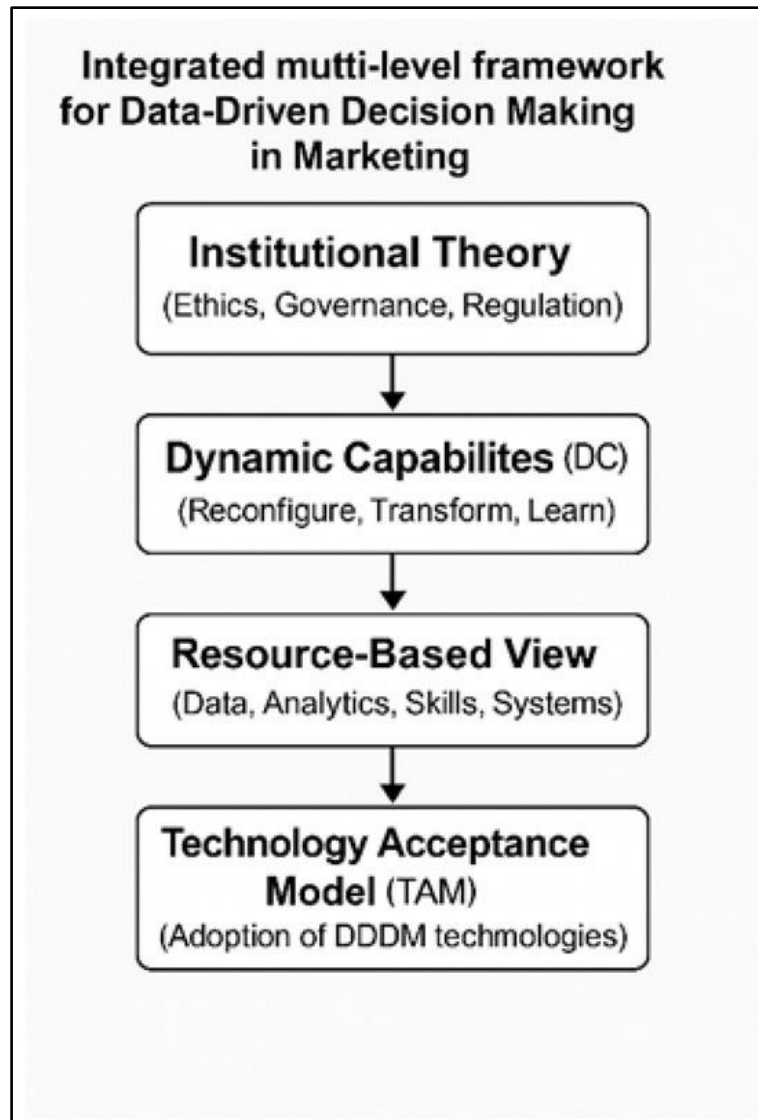


Figure 2.6: Data-Driven Decision-Making in Marketing

(Source: Cruz and Rosário, 2025)

With this theory, one can see that there is some direct connection with the aim of creating successful machine learning models to predict the success of digital marketing campaigns. Historical campaign pattern analysis can be used to build predictive models that analyze which factors are significant in campaign performance and what might be expected in the future (Sylvestre, 2024). Data driven decision making enables marketers to deploy their budgets more effectively, better target their media but also improve how their business does and how they do

their media. The theory also emphasizes the need for quality data, analytical skills, and results interpretation, when applying predictive solutions (Prakash, 2024). In this way, this framework contributes to the research by examining the role of predictive analytics and machine learning in making better marketing decisions and being more effective in increasingly data-driven digital environments.

2.7 Literature gap

Predictive analytics and machine learning are already a growing focus in the ever-evolving world of digital marketing, as discussed in existing literature. There are a few areas of research that are still lacking. Existing research has primarily focused on marketing prediction issues at the fine level of specific marketing activities instead of on the level of predicting overall digital marketing campaign successes. Furthermore, there has been limited research that encourages comparisons of different machine learning models, specifically when using the same set of data for the same conditions, to determine which one is most successful in predicting. There is also a lack of insight on what aspects of an advocacy campaign can make a difference in terms of outcomes. Hence, this study was oriented to fill these gaps by investigating the marketing performance variables, creating predictive models and assessing the different machine learning method applicability to choose the best model to forecast the success of digital marketing campaigns.

2.8 Summary

The literature review conducted in this chapter, revealed the current works covering the topics of predictive analytics, predictive machine learning and digital marketing campaign success prediction. It looked at key campaign performance elements, the utilization of predictive analytics and how an application of machine learning models can predict marketing results. Model evaluation methods were also explained and the value of using appropriate model performance metrics to compare predictive methods was discussed. Moreover, the theoretical framework included the significance of the technology adoption and data-driven decision-making perspectives. From the literature review, the research gaps identified are limited comparative studies of the use and comparative advantages between machine learning models based on the success of digital marketing campaigns.

Chapter 3: Research Methodology

3.1 Introduction

The methodology followed to investigate the use of predictive analytics for predicting the success of digital marketing campaigns is explained in this chapter. The methodology was tailored to the objectives of the research, which were: important factors of marketing performance, preparing and analysing secondary marketing data, developing machine learning models and assessment of machine learning models predictive performance. Each case includes a similar set of steps in the research process: Dataset preparation Data preprocessing Exploratory Data Analysis (EDA) Feature engineering Model development Model training and testing Performance evaluation Feature importance analysis. Campaign response is predicted using three classification algorithms: Logistic Regression, Decision Tree and Random Forest. The entire process of analysis is done using python and its data science and machine learning libraries.

3.2 Research Approach

A **quantitative research approach** is chosen because the data the researcher used for analysis were the numerical and categorical data from the customers and marketing. The quantitative analysis is indicated for this research because it will allow for Measurable relationship and pattern present in marketing information to be systematically examined (Wirtz and Balzer, 2023). These variables are then assessed with predictive analytics to see if campaign answers can be predicted with a machine learning approach. The use of a quantitative approach also helps to objectively analyse the performance of these predictive models numerically.

3.3 Research Design

The study is designed in a **predictive analytics research** manner. The process starts by acquiring and comprehending the secondary data and then followed by checking secondary data structure, data types, data descriptives, missing values and data quality. Then the data preprocessing is done, which processes the observations for analysis. Exploratory analysis helps to gain different insights about the customer characteristics, spending pattern, purchasing behavior, campaign response etc.,

in relation to different variables. Relevant additional variables are developed with feature engineering. Once the exploration is completed, the data set is split up into training and testing sets and three machine learning classification models are created. The predictions are assessed using the same performance measures which provides a consistent comparison (Sapkota *et al.*, 2026). Lastly, feature importance analysis is performed to identify those features that are most important when creating a predictive model.

3.4 Dataset Description

The data set under analysis is a secondary ***digital marketing campaign dataset*** that contains customer characteristics, purchasing activities, amount spent on products, previous campaign acceptance and whether or not a customer has responded to a particular marketing campaign. Response is the target variable to be used for the classification process in machine learning (El-Hajj and Pavlova, 2024). The value that represents a positive reaction (campaign acceptance) and another value that represents non-acceptance. The dataset had information on marketing campaign variables, as well as demographic, behavioural and spending variables that could be used for analysis of factors associated with marketing campaign response.

3.5 Data Preprocessing

Data preprocessing is the preliminary task done before exploratory analysis and machine learning to enhance data quality and make the data variables appropriate for modelling. This comprises looking at missing observations, excluding unneeded variables, changing variables to suitable data types, changing categorical data and creating new analytical features (Muraina *et al.*, 2023). While machine learning algorithms need the format in a consistent, usable manner, pre-processing has been proved especially vital. The resulting dataset is then used in EDA and for the creation of predictive models.

3.5.1 Data Cleaning

The initial step in pre-processing is to inspect the dataset to identify data quality issues. Observations and variables are verified and data types, data descriptives and duplicates of

observations and unnecessary columns are examined. A column such as an index column which will not be used effectively to predict campaign response is discarded from the analytical file (Zhang *et al.*, 2023). In addition, numerical and categorical variables are included to ensure that they are represented appropriately. These procedures eliminate irrelevant information and create a "cleaner" set of data to be analyzed later.

3.5.2 Missing Value Treatment

The numbers of missing values are computed for each variable to identify missing values. The Income variable is mentioned in particular, as income is an important variable to consider, since the lack of income data might impact the customer level analysis as well as the model construction. The missing values of the Income data are filled in by the value of the median of the Income data that are present (Alam *et al.*, 2023). After imputation, a check of the variable set involved is performed to ensure that there are no missing values, allowing for calculation of the necessary values.

3.5.3 Data Transformation

Data transformation is done to make the original variables amenable to analysis and machine learning. This Dt_Customer is transformed into a date variable to properly extract customer registration information as needed. Before machine learning modelling, numerical representations have to be used for categorical variables like Education and Marital_Status. Logistic Regression: Numerical features are standardized to avoid the difference in measurement scales having disproportionate effects on the estimation of the model (Miah *et al.*, 2024). Significantly, a scale parameters are calculated by using the training data and then applied on the testing data so that the information from the test set does not affect the preparation of the model.

3.5.4 Feature Engineering

Using feature engineering, variables are created that better summarize customer behavior. Total_Spending represents the spending on wines, fruits, meat products, fish products, sweet products and gold products. Total_Purchases is simply the sum of purchases in the Web, Catalogue

and Store channels. Campaign_Acceptance = All the former campaigns accepted. An approximation of the Customer Age based on the year of birth. These engineered variables contain aggregate information relating to customer value and purchase activity as well as past campaign interactions and can be used not only for exploratory analysis, but also to predict response for campaigns.

3.6 Exploratory Data Analysis

Exploratory Data Analysis is performed to gain insight into the nature, form, relationships and distributions in the marketing data prior to data modelling. In addition to Descriptive statistics, graphical analysis is provided to gain comprehensive understanding of customer's characteristics and marketing behaviour (Bohara, 2025). Frequencies and value(s) are added to the corresponding charts as data labels for ease of interpretation. Additionally, EDAs can be used to find potential useful predictors of campaign response.

3.6.1 Univariate Analysis

Univariate Analysis: Involves looking at one variable in isolation and to understand the distribution of that variable. The target variable Response is analysed and the number of positive and negative campaign responses is determined and potential class imbalance is found. Histograms are used to investigate distributions of customer age and income and look for unusual values (Hariguna and Chen, 2024). Product spending and purchasing channels are even summarised to see where the consumers spend and where they purchase. These analysis gives the general features of the data set before any relationships between variables are looked at.

3.6.2 Bivariate Analysis

Bivariate analysis explores the relationship between the Campaign Response and the selected Customer and Marketing characteristics. For instance, total campaign spend is analysed by the corresponding response categories to look at whether there are spending differences related to those who respond to campaigns and those who don't. The campaign response rate can also be analysed down to education and other customer levels (El-Hajj and Pavlova, 2024). The goal of

this stage is not one of cause and effect, but exploratory; the idea is to use differences that are seen to indicate patterns that might be relevant to predictive modelling.

3.6.3 Correlation Analysis

Numerical variables are analyzed using correlation analysis to discover the linear relationship between customer, spending, purchasing and campaign related variables and the nature/direction of that relationship. A matrix of correlation is computed and displayed as a heatmap. This can be used to check for monotonic relationships between variables and to select predictors with strong monotonic relationships. Correlation with the Response variable can also offer initial insights into variables related to the outcome of a campaign (Wang *et al.*, 2022). However, the correlation is being considered as association rather than a relationship of cause and effect.

3.7 Machine Learning Model Development

Three supervised machine learning classification algorithms are built: Logistic Regression, Decision Tree, and Random Forest to predict the binary campaign Response variable. When multiple algorithms are employed, the comparative tests can be conducted on the same testing data, which helps compare the predictive behaviour of the algorithms (Gupta *et al.*, 2024). Each model is trained with the predictor variables created at the preprocessing phase, and uses that model to create campaign response predictions for the test observations that it did not see during training.

3.7.1 Logistic Regression

Logistic Regression is used as a statistical classification model for estimating the probability of a binary campaign response. It gives a good baseline to assess if the available customer and market attributes are good to determine who is likely to respond positively or negatively. Standardised predictor variables are used when fitting the Logistic Regression model (Roustaei, 2024). The results of the predictions are analyzed with the same measures used for the tree-based models and predictions can be compared directly for predictive quality.

3.7.2 Decision Tree

A Decision Tree classifier is built, it has a set of decision rules which are derived from predictor variables to forecast response to campaign (El-Hajj and Pavlova, 2024). It keeps dividing the observations recursively to group them increasingly associated with some response classes, based on the feature value. An advantage of using the Decision Tree in this study is that the model can capture nonlinear relationships and interactions without having to go through the same scaling procedure as Logistic Regression.

3.7.3 Random Forest

Random Forest is used as an ensemble classification technique consisting of multiple decision trees (Jalal *et al.*, 2022). The individual trees' predictions are aggregated to yield the final classification. The method is incorporated as it can find nonlinear relationships and interactions between customer variables and marketing variables with the least requirement of one decision tree. Also, feature importance analysis is carried out using the Random Forest model, which gives importance values and can be used to rank the predictors based on its importance.

3.8 Model Training and Testing

The prepared data set is split into train and test sets (80:20) It is worth noting that 80 % of the observations are employed for training, and the remaining 20 % for evaluation. The split is done reproducibly using a fixed random state. The training subset is used for deriving parameters depending on the data, pre-processing the data, and training model fitting, but the test subset is not used during training (Mokari *et al.*, 2023). This split is more suitable for determining the performance of the models upon observations not used to estimate them.

3.9 Model Evaluation

The evaluation of model performance is done by using accuracy, precision, recall, f1-score and confusion matrix. All observations that are correctly classified is what accuracy is. Precision refers to what percentage of the positive predictions were actually positive and Recall refers to what

percentage of actual positives were correctly predicted by the model. The F1-Score is a combination of precision and recall, and it would be beneficial if we consider their balance. Several metrics are used instead of accuracy to better assess the predictive performance as campaign response classes may be unbalanced.

3.10 Feature Importance Analysis

The main model operated for feature importance analysis is the RF model. The magnitude of the importance scores that the model produces are pulled out and ordered from most important to least important, which are then displayed in a bar graph including the top predictors for importance (Inglis *et al.*, 2022). This analysis is to be used to help answer the research question of which CM and digital marketing factors contribute to the prediction of the outcomes of the campaigns. These scores are referred to as degrees of predictive relevance in the model, and not as evidence of a variable causing campaign success.

3.11 Tools and Software

The analysis is conducted through the tools available in colab notebook python. Data is manipulated, cleaned, numerically calculated and features are constructed using Pandas and NumPy (Ranjan *et al.*, 2023). Descriptive and analytical visualizations are created using Matplotlib and Seaborn, such as histograms, count plots, bar charts, box plots, and correlation heatmaps. The data splitting, feature scaling, Logistic Regression, Decision Tree, Random Forest modelling, prediction and model evaluation is done using Scikit-learn. These tools offer a single platform where you can do data preprocessing, exploratory analysis, predictive modelling and evaluation.

3.12 Ethical Considerations

The study is conducted with a secondary database instead of participants surveyed. The analysis is done for research purposes and relates to the overall characteristics and marketing actions of customers (García-Salirrosas and Rondon-Eusebio, 2022). Only variables that relate to the research question(s) are used and the results of those variables are interpreted carefully to ensure that predictive associations are not converted to causal relationships. The model results represent

just analytical insights from available data – not generalizations about all customers or digital marketing campaigns.

3.13 Chapter Summary

The methodology underlying the study of predictive analytics for the success of digital marketing campaigns has been introduced in this chapter. The research has quantitative predictive approach using secondary marketing data. The methodology involves cleaning the data, treating missing values, transforming the data, feature engineering, exploratory data analysis, and analysing the data to understand the correlation between different features prior to the predictive modelling. Logistic Regression, DecisionTree and RandomForest classifiers are trained and tested and the performance is evaluated using accuracy, precision, recall, F1-score and confusion matrices. These stages work together to give a structure to the research objectives to be able to support the presentation and discussion of empirical findings.

Chapter 4: Findings Analysis and Discussion

4.1 Introduction

In this chapter, the learner discusses data analysis, results of the process of implementing predictive analytics and machine learning algorithms to predict the success of digital marketing campaigns. It is designed and analyzed using the methodology presented in Chapter #3 and should strive to analyze trends in customer behavior, assess which marketing factors affect behavior and identify the effectiveness of predictive models. The chapter starts with a description of the dataset and descriptive analysis, then the result of data preprocessing, the result of exploratory data analysis and the result of feature engineering. Classification performance measures, such as accuracy, precision, recall, and F1-score are applied to development and evaluation of machine learning models (Logistic Regression, Decision Tree and Random Forest). Furthermore, a feature analysis will be conducted to see what are the important features to predict the success of campaigns. Research objectives and questions are contextualized based on the results and the input of predictive analytics analysis for making the decisions in digital marketing.

4.2 Dataset Overview and Descriptive Analysis

Import Required Libraries

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
sns.set_style("whitegrid")

```

Load Marketing Campaign Dataset

```

campaign_data = pd.read_csv("marketing_campaign.csv")
campaign_data.head()

```

Unnamed: 0	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	DT_Customer	Recency	...	NumbevisitsMonth	AcceptedCmp1	AcceptedCmp4	AcceptedCmp5	AcceptedCmp6	AcceptedCmp7	Complain	Z_CostContact	Z_Revenue	Response	
0	0	5524	1957	Graduation	Single	58138.0	0	0	2012-09-04	58	...	7	0	0	0	0	0	0	3	11	1
1	1	2174	1954	Graduation	Single	46344.0	1	1	2014-03-08	38	...	5	0	0	0	0	0	0	3	11	0
2	2	4141	1965	Graduation	Together	71613.0	0	0	2013-08-21	26	...	4	0	0	0	0	0	0	3	11	0
3	3	6182	1984	Graduation	Together	26646.0	1	0	2014-02-10	26	...	6	0	0	0	0	0	0	3	11	0
4	4	5324	1981	PhD	Married	58293.0	1	0	2014-01-19	94	...	5	0	0	0	0	0	0	3	11	0

5 rows x 30 columns

Figure 4.1: Loading of Digital Marketing Campaign Dataset

figure 4.1 shows the first loading of the marketing campaign database, into Python environment. The variables included in the dataset are the demographics of the customers, their buying behaviour, their response to the campaigns and the variables that are related to marketing



interactions. This information is used as a foundation for the digital marketing campaign's effectiveness and/or success.

Dataset Structure and Information

```
[3] print("Dataset Shape:", campaign_data.shape)
✓ Os campaign_data.info()

... Dataset Shape: (2240, 30)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 30 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Unnamed: 0                            2240 non-null   int64
1   ID                                     2240 non-null   int64
2   Year_Birth                            2240 non-null   int64
3   Education                             2240 non-null   object
4   Marital_Status                        2240 non-null   object
5   Income                                2216 non-null   float64
6   Kidhome                               2240 non-null   int64
7   Teenhome                              2240 non-null   int64
8   Dt_Customer                           2240 non-null   object
9   Recency                               2240 non-null   int64
10  MntWines                              2240 non-null   int64
11  MntFruits                             2240 non-null   int64
12  MntMeatProducts                       2240 non-null   int64
13  MntFishProducts                       2240 non-null   int64
14  MntSweetProducts                      2240 non-null   int64
15  MntGoldProds                          2240 non-null   int64
16  NumDealsPurchases                     2240 non-null   int64
17  NumWebPurchases                       2240 non-null   int64
18  NumCatalogPurchases                   2240 non-null   int64
19  NumStorePurchases                     2240 non-null   int64
20  NumWebVisitsMonth                     2240 non-null   int64
21  AcceptedCmp3                          2240 non-null   int64
22  AcceptedCmp4                          2240 non-null   int64
23  AcceptedCmp5                          2240 non-null   int64
24  AcceptedCmp1                          2240 non-null   int64
25  AcceptedCmp2                          2240 non-null   int64
26  Complain                              2240 non-null   int64
27  Z_CostContact                          2240 non-null   int64
28  Z_Revenue                             2240 non-null   int64
29  Response                              2240 non-null   int64
dtypes: float64(1), int64(26), object(3)
memory usage: 525.1+ KB
```

Figure 4.2: Dataset Structure and Information Summary

Figure 4.2 shows the shape of the data if there is a massive amount of records, variables, data type and missing value within the dataset. It will consist of 2240 rows of observations and 30 attributes, both numerical and non-numerical, which will be related to the customers and output of the activities carried out in the campaign.

Statistical Summary of Dataset

```
campaign_data.describe()
```

	Unnamed: 0	ID	Year_Birth	Income	Kidhome	Teenhome	Recency	Hntbuines	HntFruits	HntHeatProducts	...	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	AcceptedCmp2	Complain
count	2240.000000	2240.000000	2240.000000	2216.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	...	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000
mean	1119.500000	5592.159821	1968.805804	52247.251354	0.444196	0.506250	49.109375	303.935714	26.302232	166.950000	...	5.316518	0.072768	0.074554	0.072768	0.064286	0.013393	0.009375
std	646.778623	3246.862198	11.984069	25173.076861	0.538398	0.544538	28.962453	336.597393	39.773434	225.715373	...	2.428645	0.259813	0.262728	0.259813	0.245316	0.114976	0.096391
min	0.000000	0.000000	1893.000000	1730.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	...	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	559.750000	2828.250000	1959.000000	35303.000000	0.000000	0.000000	24.000000	23.750000	1.000000	16.000000	...	3.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	1119.500000	5458.500000	1970.000000	51381.500000	0.000000	0.000000	49.000000	173.500000	8.000000	67.000000	...	6.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	1679.250000	8427.750000	1977.000000	68522.000000	1.000000	1.000000	74.000000	504.250000	33.000000	232.000000	...	7.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
max	2239.000000	11191.000000	1996.000000	666666.000000	2.000000	2.000000	99.000000	1493.000000	199.000000	1725.000000	...	20.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000

8 rows x 27 columns

Figure 4.3: Statistical Summary of Marketing Campaign Dataset

Descriptive statistics of numerical variables, such as count, mean, standard deviation, minimum, maximum and quartile values are also presented in figure 4.3. The differences in the customer characteristics, buying behaviour, percentage of campaign acceptance and reaction behaviour are discussed in the summary, leading up to extra exploratory data analysis.



...

	Missing Values	Percentage
Income	24	1.071429
Unnamed: 0	0	0.000000
ID	0	0.000000
Year_Birth	0	0.000000
Education	0	0.000000
Marital_Status	0	0.000000
Kidhome	0	0.000000
Teenhome	0	0.000000
Dt_Customer	0	0.000000
Recency	0	0.000000
MntWines	0	0.000000
MntFruits	0	0.000000
MntMeatProducts	0	0.000000
MntFishProducts	0	0.000000
MntSweetProducts	0	0.000000
MntGoldProds	0	0.000000
NumDealsPurchases	0	0.000000
NumWebPurchases	0	0.000000
NumCatalogPurchases	0	0.000000
NumStorePurchases	0	0.000000
NumWebVisitsMonth	0	0.000000
AcceptedCmp3	0	0.000000
AcceptedCmp4	0	0.000000
AcceptedCmp5	0	0.000000
AcceptedCmp1	0	0.000000
AcceptedCmp2	0	0.000000
Complain	0	0.000000
Z_CostContact	0	0.000000
Z_Revenue	0	0.000000
Response	0	0.000000

Figure 4.4: Missing Values Analysis of Marketing Campaign Dataset

Figure 4.4, shows the analysis of the missing values of the data set. It is revealed that most of the variables have all records, there are 24 missing values of the Income variable. Such an evaluation aids in data preprocessing choices, associated with building sound machine learning models.

4.3 Data Preprocessing Results

```

v Convert Customer Registration Date

[7] ✓ 0s
campaign_data["Dt_Customer"] = pd.to_datetime(campaign_data["Dt_Customer"])
campaign_data["Customer_Year"] = campaign_data["Dt_Customer"].dt.year
campaign_data["Customer_Month"] = campaign_data["Dt_Customer"].dt.month

Feature Engineering

v Total Product Spending

[8] ✓ 0s
campaign_data["Total_Spending"] = (campaign_data["MntWines"]+campaign_data["MntFruits"]+campaign_data["MntVeatProducts"]+campaign_data["MntFishProducts"]+campaign_data["MntSweetProducts"]+campaign_data["MntGoldProds"])

v Total Purchase Behaviour

[9] ✓ 0s
campaign_data["Total_Purchases"] = (campaign_data["NumWebPurchases"]+campaign_data["NumCatalogPurchases"]+campaign_data["NumStorePurchases"])

v Previous Campaign Acceptance

[10] ✓ 0s
campaign_data["Campaign_Acceptance"] = (campaign_data["AcceptedCmp1"]+campaign_data["AcceptedCmp2"]+campaign_data["AcceptedCmp3"]+campaign_data["AcceptedCmp4"]+campaign_data["AcceptedCmp5"])

```

Figure 4.5: Feature Engineering Process for Marketing Dataset

This figure shows the transformation of the unstructured customer attributes to meaningful predictive attributes. This process generates such variables as the year/month of customer, and the total amount spent, total purchases and past campaign acceptance. The engineered features gained enhance the model insight on the behaviour of the customers. The sensitivity to variability across groups is manifested in the behavioural pattern; the safety is on the properties that aid more reliable predictive judgements.

✓ Encode Categorical Variables

```
[23] ✓ Os encoder=LabelEncoder()  
categorical_columns=["Education","Marital_Status"]  
for column in categorical_columns:  
    campaign_data[column]=encoder.fit_transform(campaign_data[column])
```

✓ Define Features and Target Variable

```
[24] ✓ Os X=campaign_data.drop("Response",axis=1)  
y=campaign_data["Response"]  
print("Feature Shape:",X.shape)  
print("Target Shape:",y.shape)
```

... Feature Shape: (2240, 35)
Target Shape: (2240,)

Figure 4.6: Encoding Categorical Variables and Defining Target Variable

This code demonstrates how to prepare categorical customer variables, and inflect input features and the prediction target. The Distribution table for the data gives the details on how the population is distributed according to qualifications in education and marital status is represented as numeric codes and Response is selected as a target variable. The resulting feature of 2,240 records and 35 variables matrix. The difference between groups (groups minus groups) is mathematically represented, and this necessitates careful interpretation of the mathematical representation in order not to make biased predictions.

✓ Train-Test Split

```
[25]
✓ Os X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

✓ Feature Scaling

```
[26]
✓ Os scaler=StandardScaler()
X_train_scaled=scaler.fit_transform(X_train.drop(['Dt_Customer'],axis=1))
X_test_scaled=scaler.transform(X_test.drop(['Dt_Customer'],axis=1))
```

Figure 4.7: Train-Test Split and Feature Scaling Process

The formatting of the data is done before the training of machine learning models in this code. Training and test groups divided into 80/20 and numeric values made the same scale with the help of scaling. The resultant process enhances comparing and improving the reliability of models (Haque *et al.*, 2024). The data has customer group differences and therefore, it needs to be evaluated to make the same predictions among groups.

4.4 Exploratory Data Analysis (EDA)

✓ Customer Age Creation

```
[11]
✓ Os campaign_data["Age"] = 2025 - campaign_data["Year_Birth"]
campaign_data[["Year_Birth","Age"]].head()
```

	Year_Birth	Age
0	1957	68
1	1954	71
2	1965	60
3	1984	41
4	1981	44

Figure 4.8: Customer Age Feature Creation

The metric takes the year of birth and transforms it to customer age to use in demographic analysis. The examples of customer ages between 41 and 71 years are computed. Age groups depict the dissimilarities in attributes of customers and its purchasing behavior. Despite the age factor potentially affecting predictions, its application should be done with a lot of care since applying demographic factors directly can create unfair disparages in automated decision making processes.

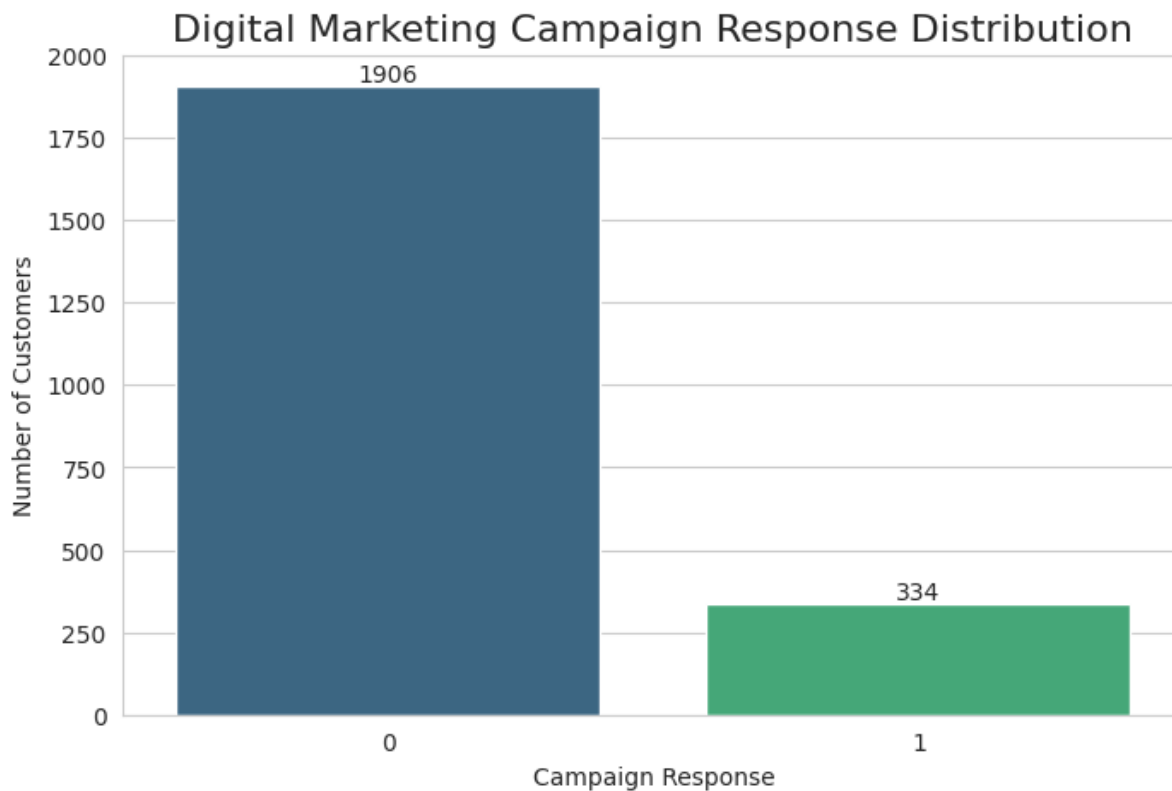


Figure 4.9: Digital Marketing Campaign Response Distribution

This value is used to quantify the spread of customer feedbacks to the marketing campaign. The results obtained give a numbers of 1,906 customers who did not respond and 334 customers who responded positively. The groups are significantly different and hence, there is an unbalanced target variable. In predictive decisions, the difference implies that models have to consider imbalance to prevent biases towards the majority group and lead to biased customer classification results.

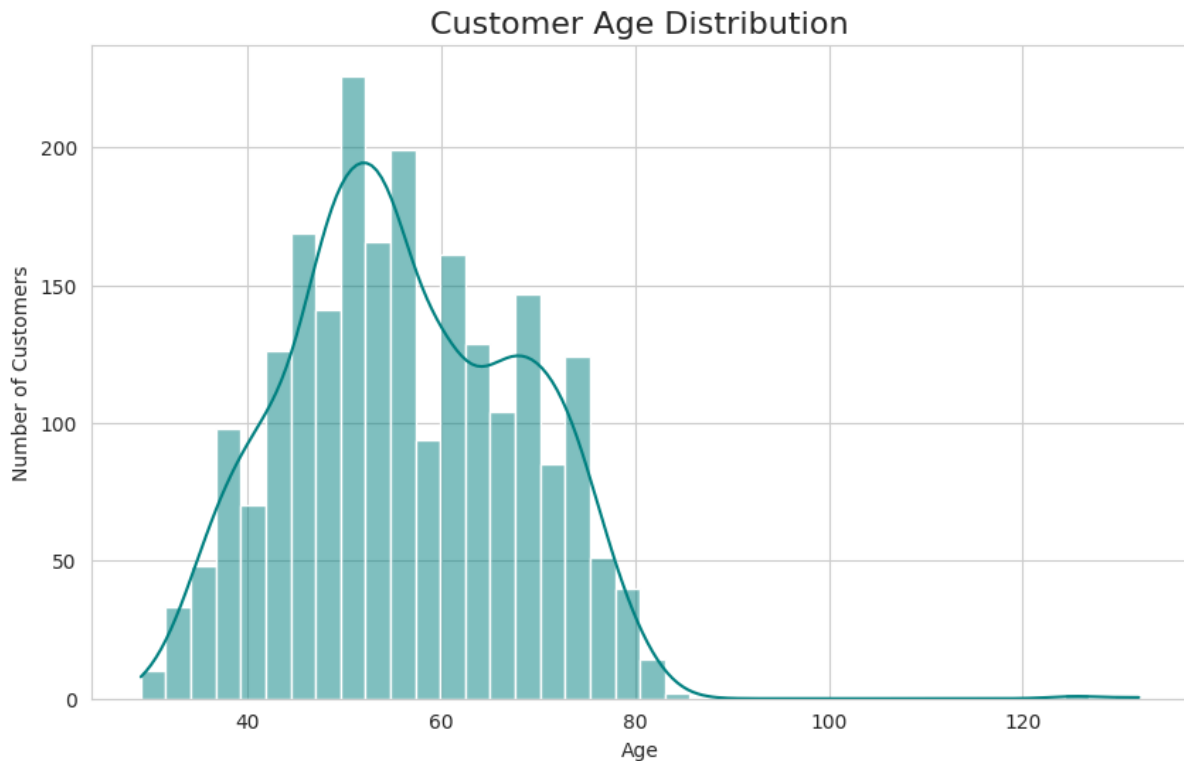


Figure 4.10: Customer Age Distribution

The metric gauges the occurrence rate of the ages of customers in the dataset. The findings show that the bulk of the customers lies in the range of about 40-70 years. The representation varies among age groups with middle age customers representing the biggest group (Miah and Islam, 2024). This variation suggests that age behaviour also must be taken into account when making automated decisions and that the general age trend should be tracked to encourage predictions that are not a result of demographic bias.

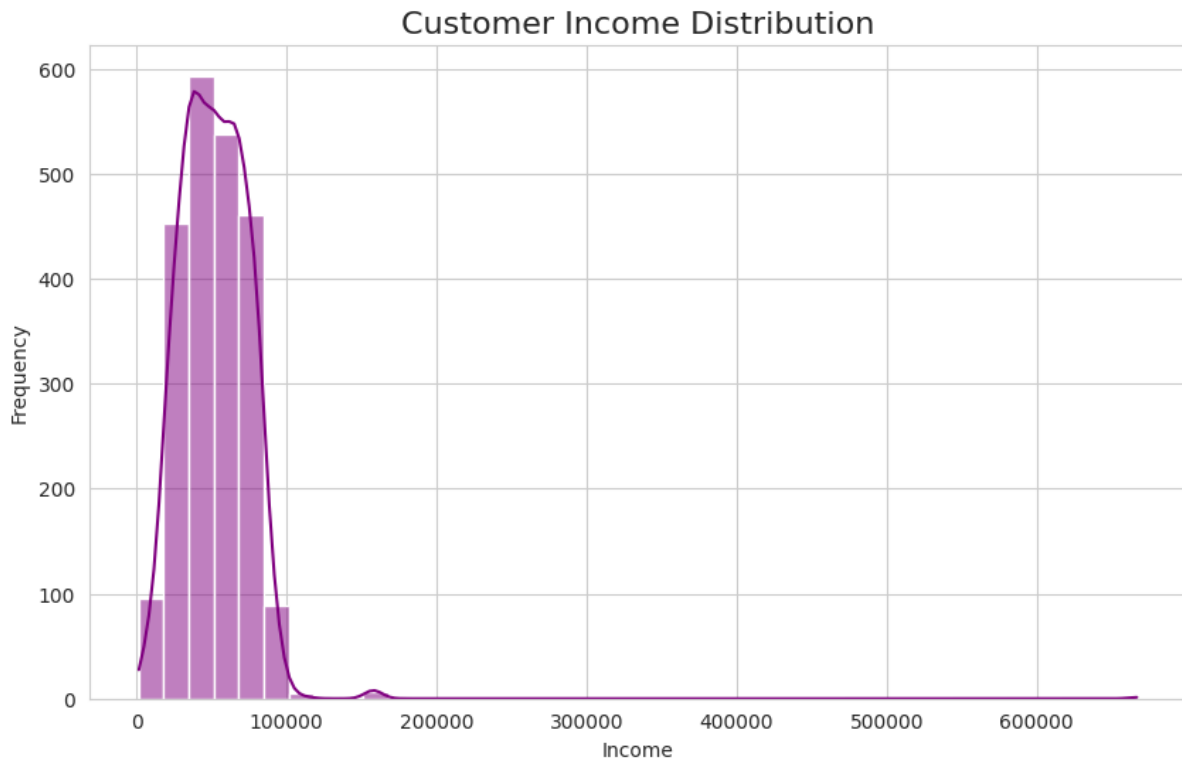


Figure 4.11: Customer Income Distribution

This measure considers the distribution of the values of customer incomes in the dataset. The results revealed that the majority of consumers earn between c. 0 to 100,000 with a few scattered around being able to earn more. There is a marked disparity in terms of representation that has an effect on income groups. Care must be taken to not allow high-income groups to dominate the predictive consequences of these differences, if they are to be used for decision making systems.

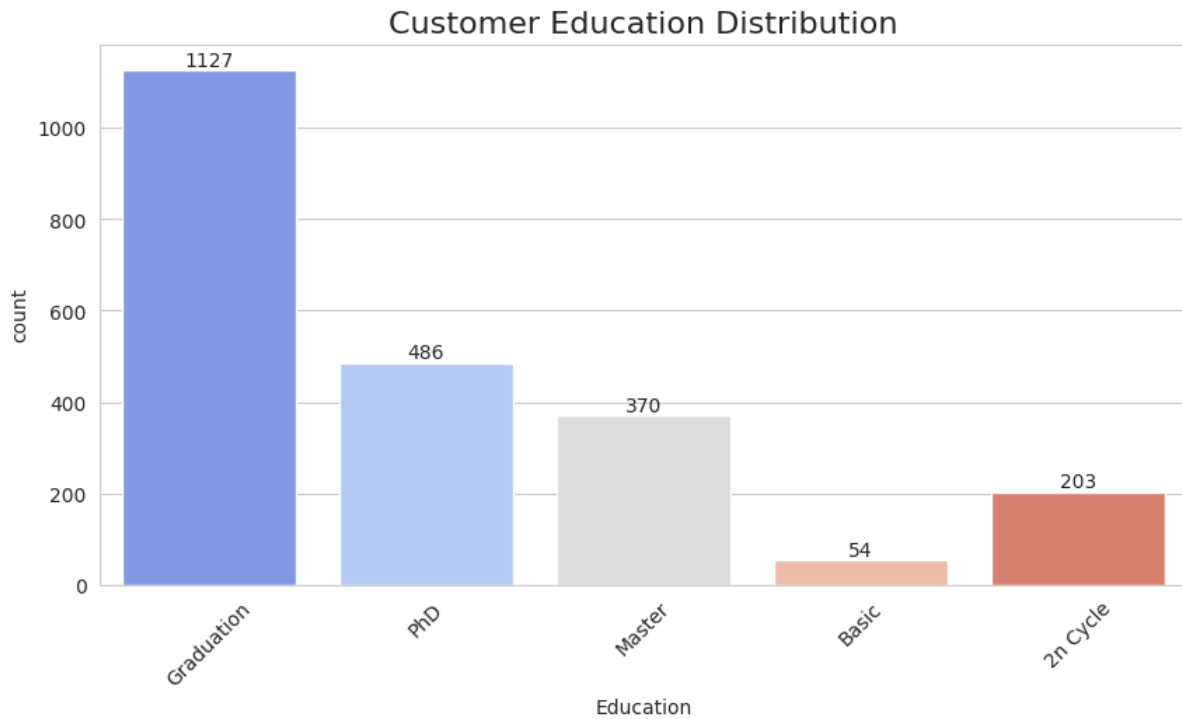


Figure 4.12: Customer Education Distribution

This value is an indicator of the customer distribution of the various levels of education. The results show that Basic, 2n Cycle, Master followed by PhD, are the customers with less number than the customer count of Graduation with 1127 customers. There are various types of educational groups, some are more varied than others, which may have an impact on the learning patterns of the model. Education in predictive decisions ought to be viewed with caution to prevent being unjust due to any demographic traits.

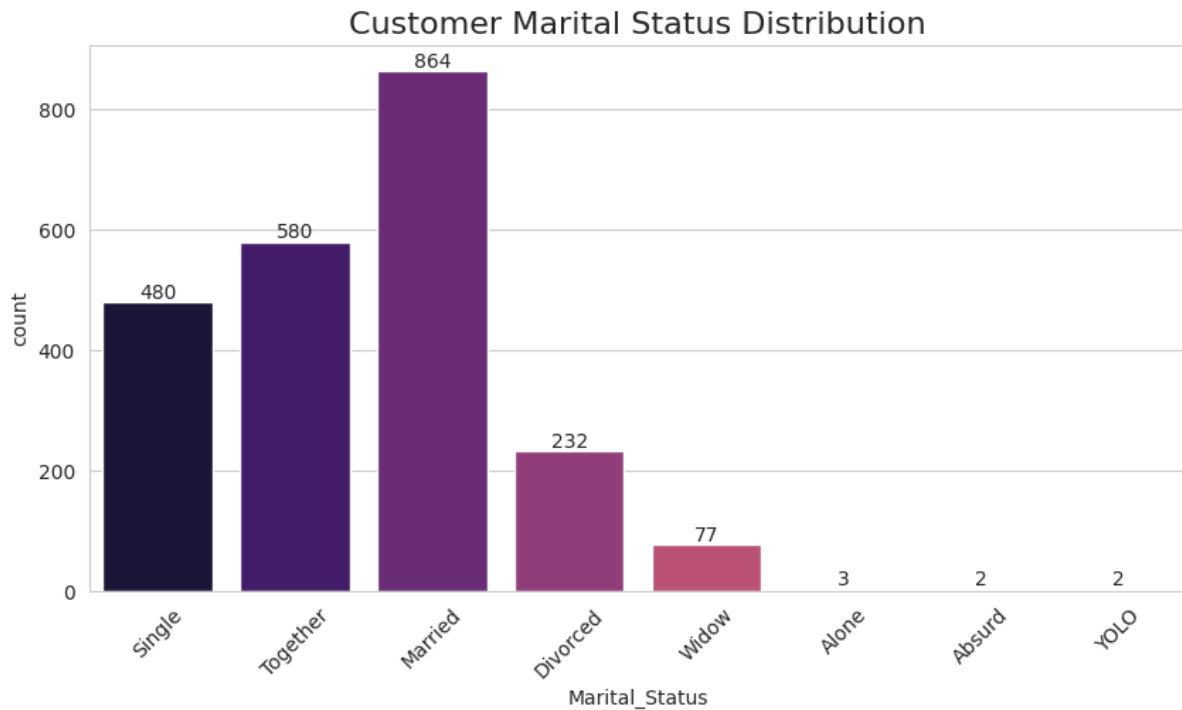


Figure 4.13: Customer Marital Status Distribution

This measure is a customer distribution measure with respect to marital status. The possible values for the number of values obtained show that the largest number of customers is Married (864) then Together and then Single. It is not fair and the minor ones (like Alone, Absurd and YOLO) are omitted a lot. Unequal group size can also impact a model of predictive decision-making and should be assessed, to have balanced results.

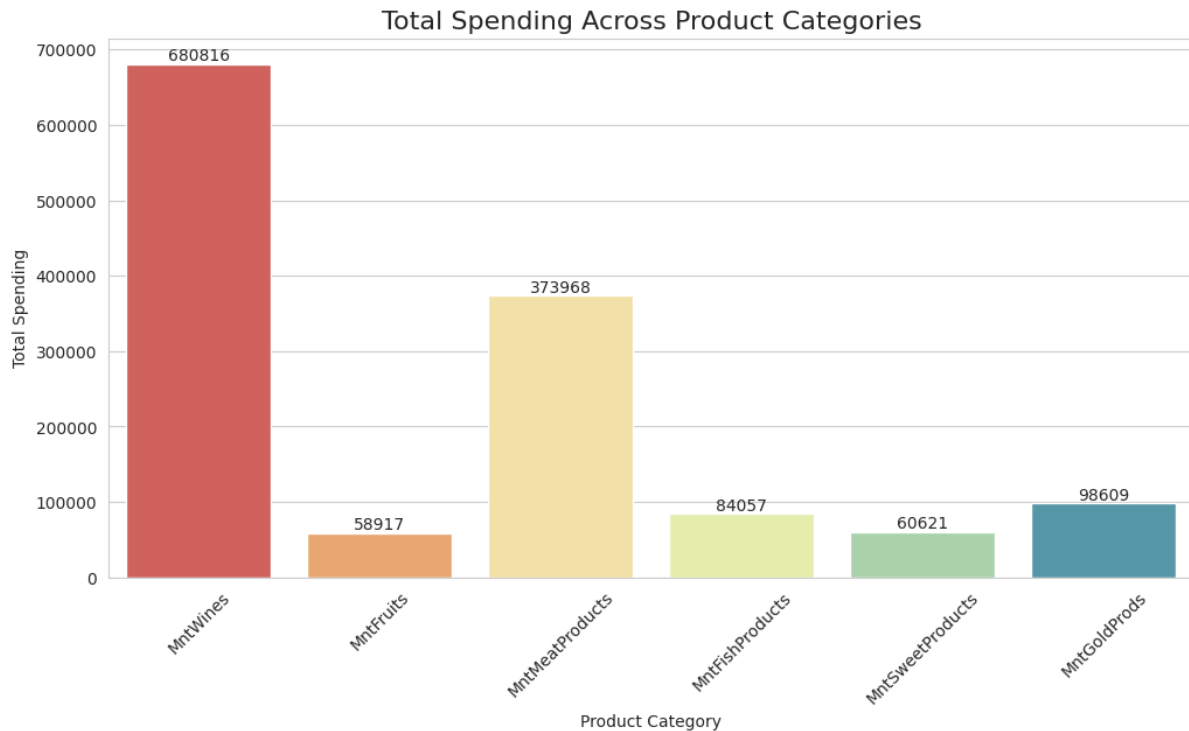


Figure 4.14: Total Spending Across Product Categories

This number is a summation of the amount of money customers are spending in various product lines. The results have revealed that the total expenditure on wine products is the highest of Rs. 680,816 and total expenditure on meat products is Rs. 373,968. Other categories are of lower values. Customers can be divided into groups according to the purchasing preferences and not the safeguarded traits. Only marketing decisions could be made based on the patterns; and the pattern must ensure that there is no unfair customer segmentation.

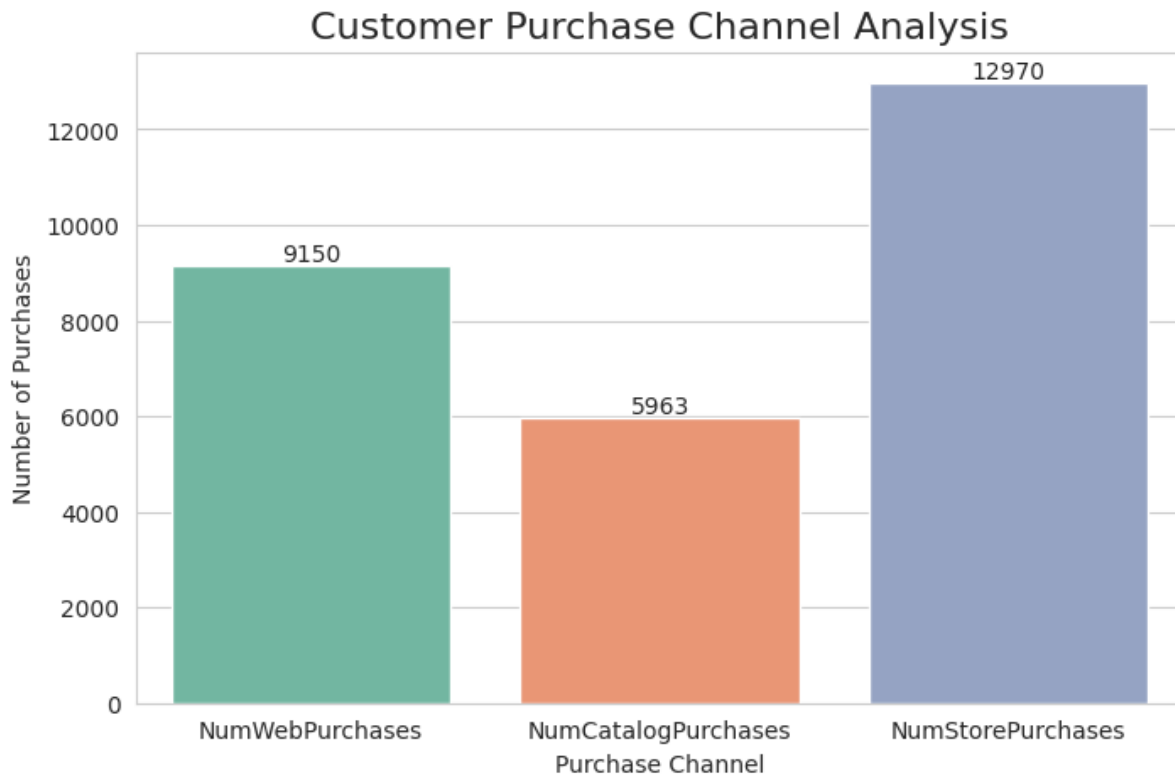


Figure 4.15: Customer Purchase Channel Analysis

An indicator of shopping patterns across the Internet, through a catalogue, as well as in-store. The outcomes show that store purchases are the most common amounting to 12,970 purchases, followed by web purchases and catalogue purchases. The differences between the groups of customers show the preferences of the channels. In predictive decisions, these behavioural differences can enhance the recommendations but this should be isolated with attributions that are sensitive to ensure that the results of the automated decisions are fair.

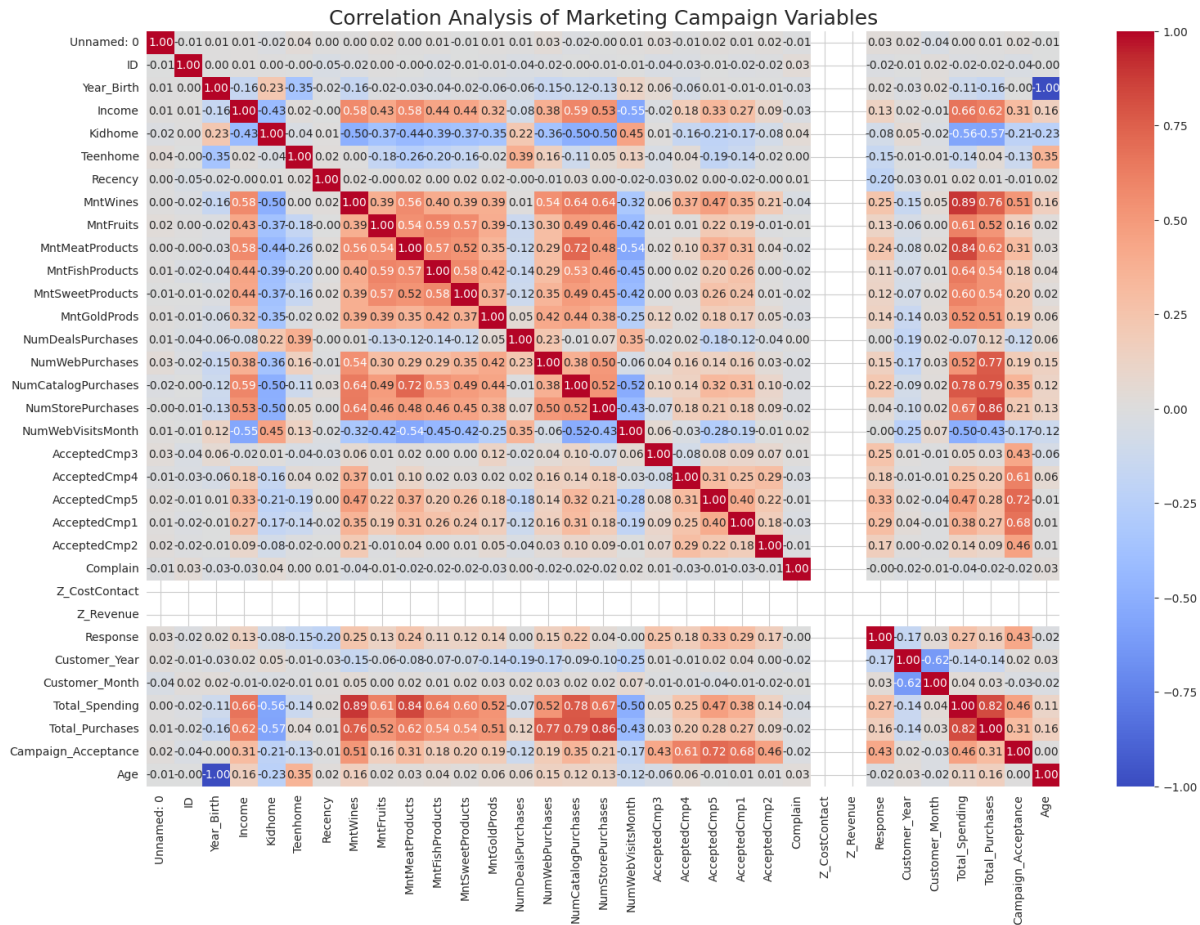


Figure 4.16: Correlation Analysis of Marketing Campaign Variables

This result is a figure that quantifies associations among marketing variables in terms of correlation coefficients means of -1 to +1. Follow strong positive and strong negative relationships, conditions that would lead to influential factors in the outcomes of the campaign. Spending, purchases, and acceptance of campaigns are shown to be significantly related to the variables (Faraz and Anjum, 2025). The differences within groups are in terms of behavioural patterns as opposed to safeguarded traits. These lessons can be used to enhance the level of prediction and minimize the unwarranted bias.

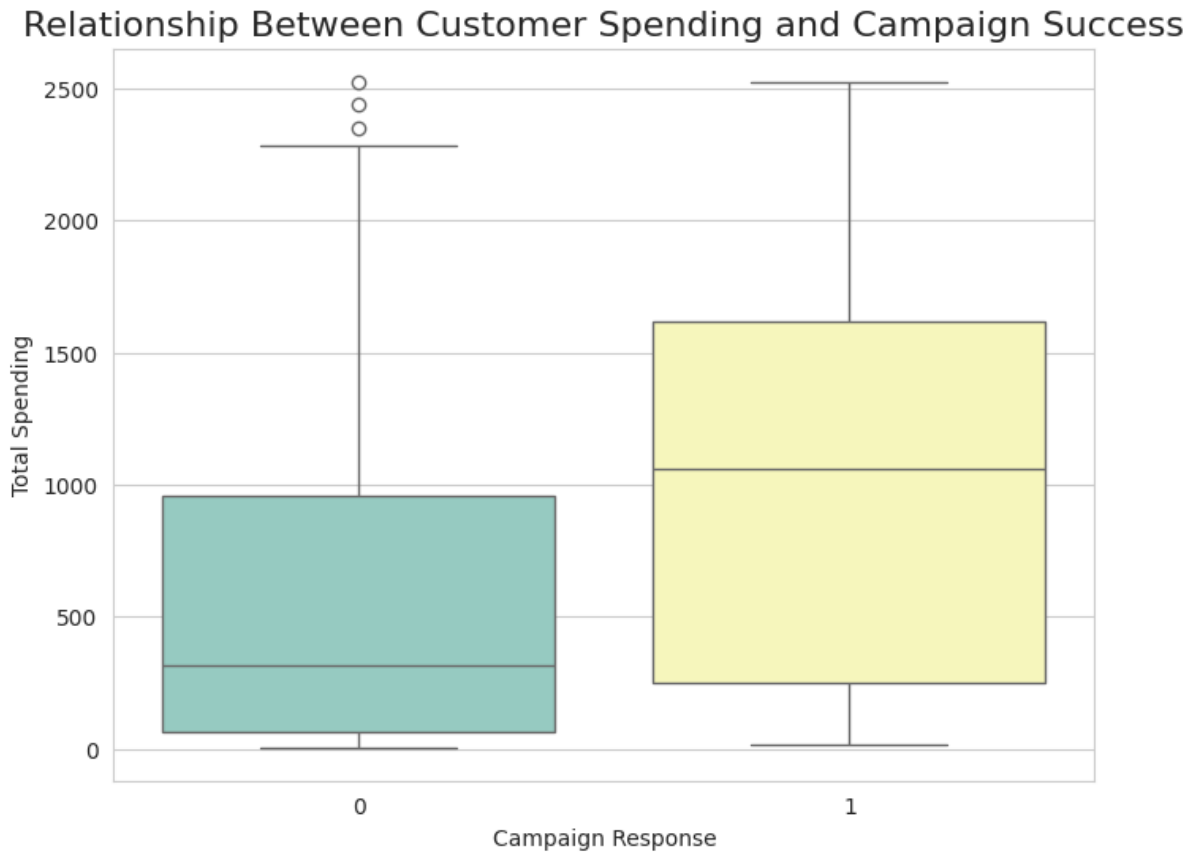


Figure 4.17: Relationship Between Customer Spending and Campaign Success

This measure is applied to compare the general level of spending by the customers and the campaign response. The boxplot shows that the median amount that successful campaign responders (Response = 1) spent is higher than the median for non-responders (Response = 0). Customers are divided into groups according to the behaviour of purchases. In the case of automated decisions, it means that it can be used to predict the spending patterns, the decision made does not have to openly discriminate against low-spending customers.

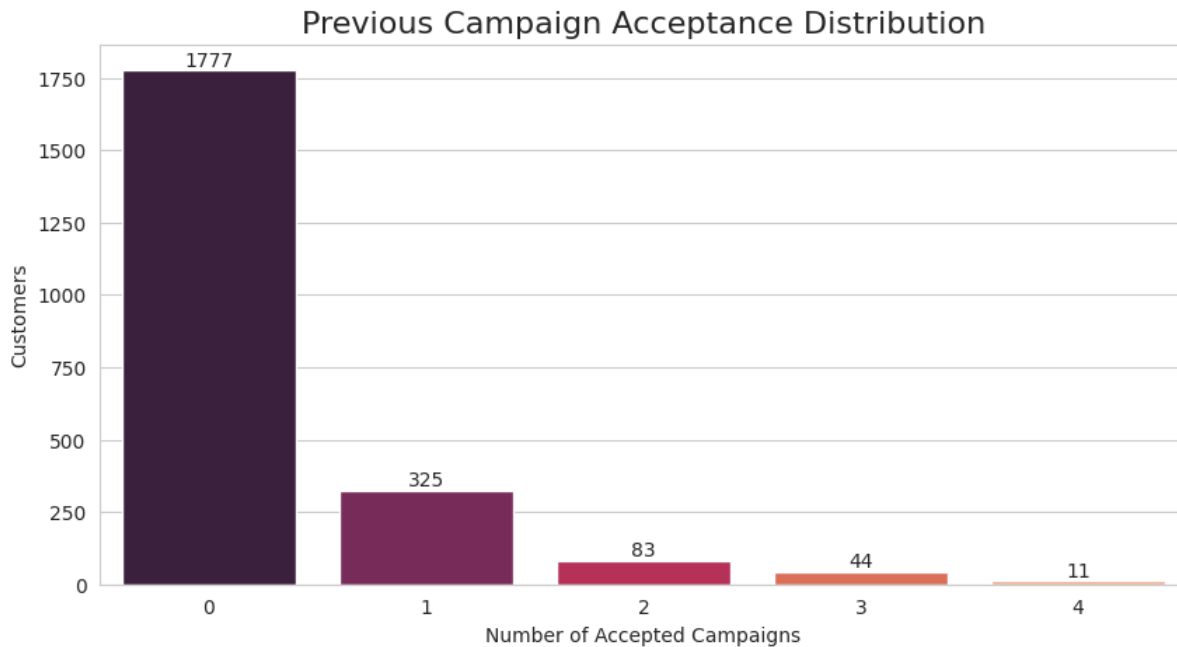


Figure 4.18: Previous Campaign Acceptance Distribution

Measures the level of campaign acceptance from the marketer's previous campaign. The values obtained indicate that 1, 777 customers did not receive any past campaigns whereas fewer customers received other campaigns. Groups vary and are divided by the extent to which they had been involved in the past. For predictive decision system, that means that the history of the response could be a useful indicator of customer activity but shouldn't be considered as the only rule to apply when designing targeting rules in the future.

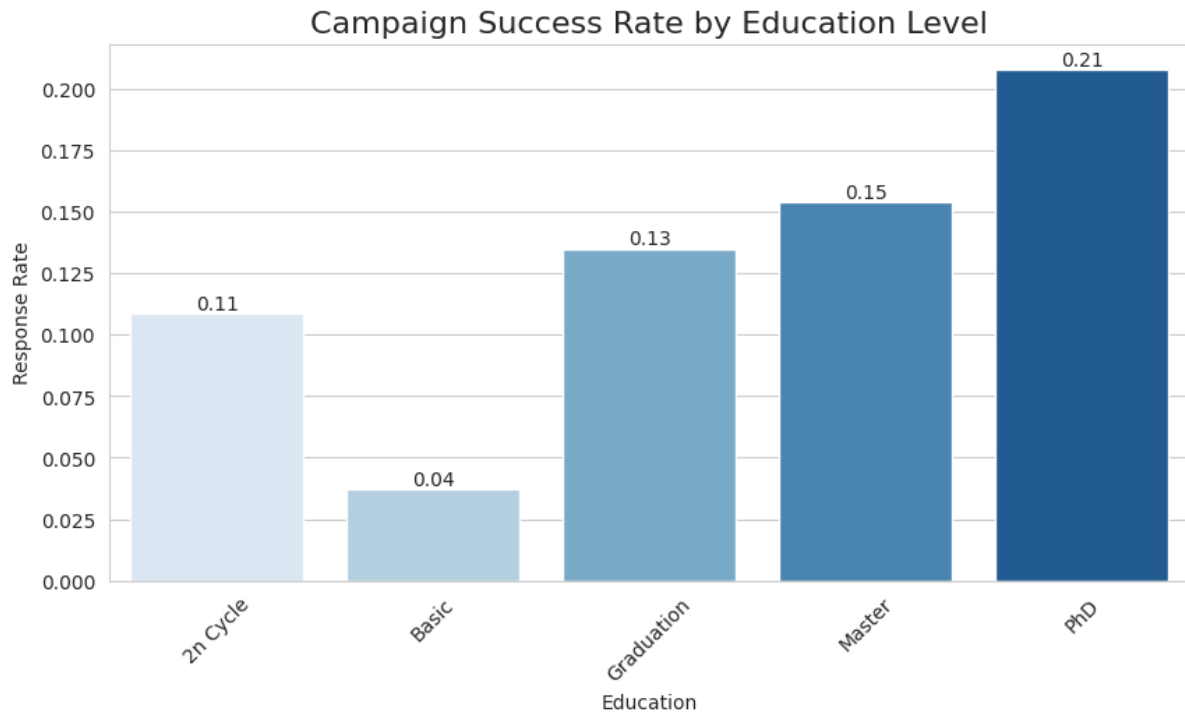


Figure 4.19: Campaign Success Rate by Education Level

This metric is used to determine the response rates of campaign of various education groups. The Customer value obtained are put PhD customer having greatest value of 0.21, Master customer value is of 0.15, Graduation value is of 0.13, 2 n Cycle (0.11) and Basic (0.04) respectively. Depending on the circumstances, teams tend to be more active or passive. If it is automated, differences associated with education should be captured through a tracking of the implications of demographic differences to remove the risk of unequal marketing opportunities.

4.5 Machine Learning Model Development and Results

✓ Build Logistic Regression Model

```
[28] ✓ 0s  
logistic_model=LogisticRegression(random_state=42,max_iter=1000)  
logistic_model.fit(X_train_scaled,y_train)  
logistic_prediction=logistic_model.predict(X_test_scaled)
```

✓ Evaluate Logistic Regression Performance

```
[29] ✓ 0s  
logistic_accuracy=accuracy_score(y_test,logistic_prediction)  
logistic_precision=precision_score(y_test,logistic_prediction)  
logistic_recall=recall_score(y_test,logistic_prediction)  
logistic_f1=f1_score(y_test,logistic_prediction)  
  
print("Logistic Regression Performance")  
print("-----")  
print("Accuracy:",logistic_accuracy)  
print("Precision:",logistic_precision)  
print("Recall:",logistic_recall)  
print("F1 Score:",logistic_f1)
```

```
✓ ... Logistic Regression Performance  
-----  
Accuracy: 0.875  
Precision: 0.6857142857142857  
Recall: 0.34782608695652173  
F1 Score: 0.46153846153846156
```

Figure 4.20: Logistic Regression Model Performance Evaluation

This measure is related to the prediction power of the Logistic Regression classification model, and it measures accuracy, precision, recall and F1-score. The obtained values are accuracy 0.875, precision 0.686, recall 0.348, and F1-score 0.462. The model performs well on correctly categorising the majority of customers, but it does not have as many positive responses to campaigns. The difference between the customer groups may be through the way they behave, but attributes related to protection are not comparatively measured (Kim *et al.*, 2024). In case of decision systems, it implies that the predictions must aid their marketing choices, without causing discrimination between various groups of customers.

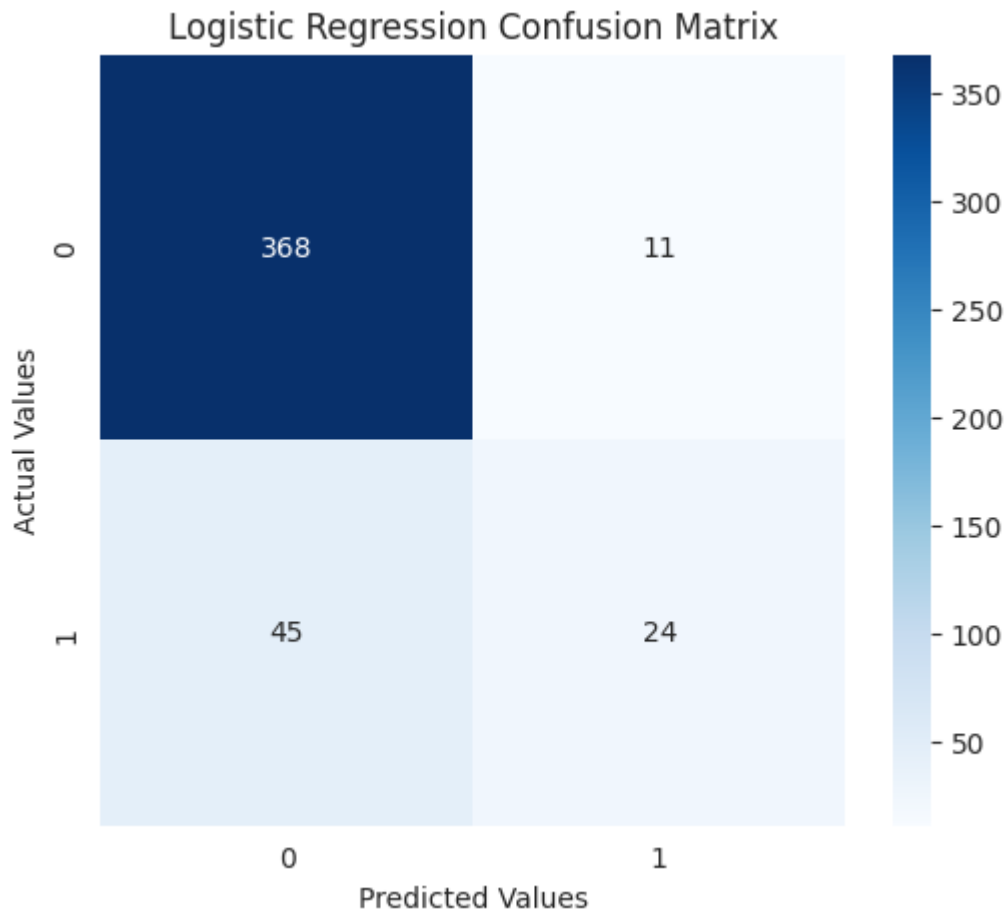


Figure 4.21: Logistic Regression Confusion Matrix

The measure is the amount of true predictions and false predictions made by the Logistic Regression model. The values derived are as true negatives = 368, true positives = 24, false negatives = 45 and a false positive = 11. This model has been more effective in defining non responsive customers as compared to respondents to successful campaigns. Intergroup differences are not associated with values but instead they pertain to behaviour of response. In the case of automated decisions: balancing should be done in order to deliver the assessment while maintaining the awareness of the customers that are likely to be receptive.

✓ Build Decision Tree Model

```
[32] ✓ Os
decision_tree_model=DecisionTreeClassifier(random_state=42)
decision_tree_model.fit(X_train.drop(['Dt_Customer'],axis=1),y_train)
decision_tree_prediction=decision_tree_model.predict(X_test.drop(['Dt_Customer'],axis=1))
```

✓ Decision Tree Evaluation

```
[33] ✓ Os
dt_accuracy=accuracy_score(y_test,decision_tree_prediction)
dt_precision=precision_score(y_test,decision_tree_prediction)
dt_recall=recall_score(y_test,decision_tree_prediction)
dt_f1=f1_score(y_test,decision_tree_prediction)
print("Decision Tree Results")
print("Accuracy:",dt_accuracy)
print("Precision:",dt_precision)
print("Recall:",dt_recall)
print("F1 Score:",dt_f1)
```

```
✓ Decision Tree Results
Accuracy: 0.796875
Precision: 0.358974358974359
Recall: 0.4057971014492754
F1 Score: 0.38095238095238093
```

Figure 4.22: Decision Tree Model Performance Evaluation

Classification accuracy, precision and recall, Decision Tree model F1-score are measures of this. The obtained values are accuracy 0.797, precision 0.359, recall 0.406, and F1-score 0.381. The overall performance of the model is less as compared to Logistic Regression. The various customer groups go to these different sites or show their own behaviour—so their prediction patterns will be different. If the model is to be used in automated decision making, the need arises to carefully validate the model, such that the customer groups get the same and unbiased output results.

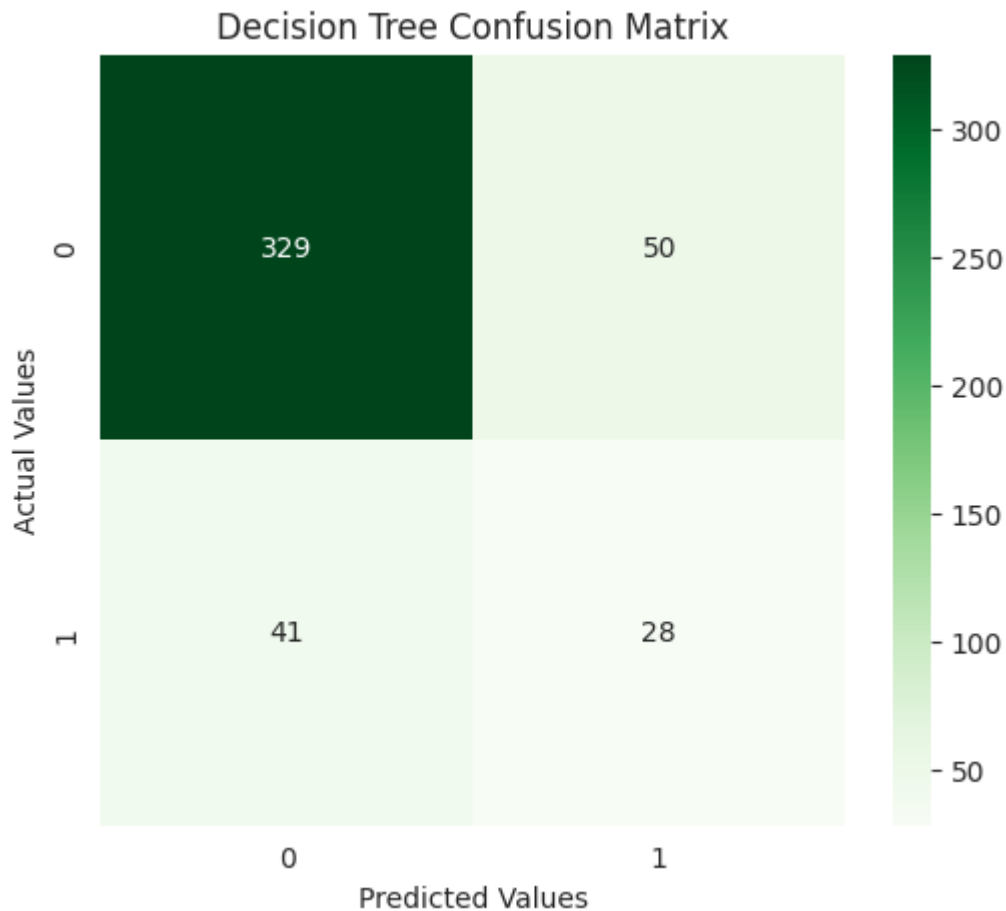


Figure 4.23: Decision Tree Confusion Matrix

This is a metric of distribution of correct and incorrect classifications produced by the Decision Tree model. The number of three values are 329, 28, 41 and 50 respectively. A larger number of positive responses can be identified by the model compared to others but incorrect positive predictions are made as well. These are the patterns of customer behaviour – the group differences. In the case of automated decision making, this underlines the use of controlling the false prediction aspects before using model based recommendations.

▼ Random Forest Model Development

```
[35] ✓ 0s random_forest_model=RandomForestClassifier(n_estimators=100,random_state=42)
random_forest_model.fit(X_train.drop(['Dt_Customer'],axis=1),y_train)
random_forest_prediction=random_forest_model.predict(X_test.drop(['Dt_Customer'],a:
```

▼ Random Forest Evaluation

```
[36] ✓ 0s rf_accuracy=accuracy_score(y_test,random_forest_prediction)
rf_precision=precision_score(y_test,random_forest_prediction)
rf_recall=recall_score(y_test,random_forest_prediction)
rf_f1=f1_score(y_test,random_forest_prediction)
print("Random Forest Results")
print("Accuracy:",rf_accuracy)
print("Precision:",rf_precision)
print("Recall:",rf_recall)
print("F1 Score:",rf_f1)
```

```
▼ *** Random Forest Results
Accuracy: 0.8660714285714286
Precision: 0.6666666666666666
Recall: 0.2608695652173913
F1 Score: 0.375
```

Figure 4.24: Random Forest Model Performance Evaluation

These results are used for the performance of the Classification model (Rand Forest) to check the accuracy of the classification model. The obtained values are accuracy 0.866, precision 0.667, recall 0.261, and F1-score 0.375. The model is highly accurate as a whole with a lower number of positive campaign responses detected. Distinctions among the groups of customers are founded on buying and activity behaviour as opposed to safeguarded traits. For automated decision making, this means that the model's prediction must be monitored to ensure that it does not adversely impact on smaller groups of customers.

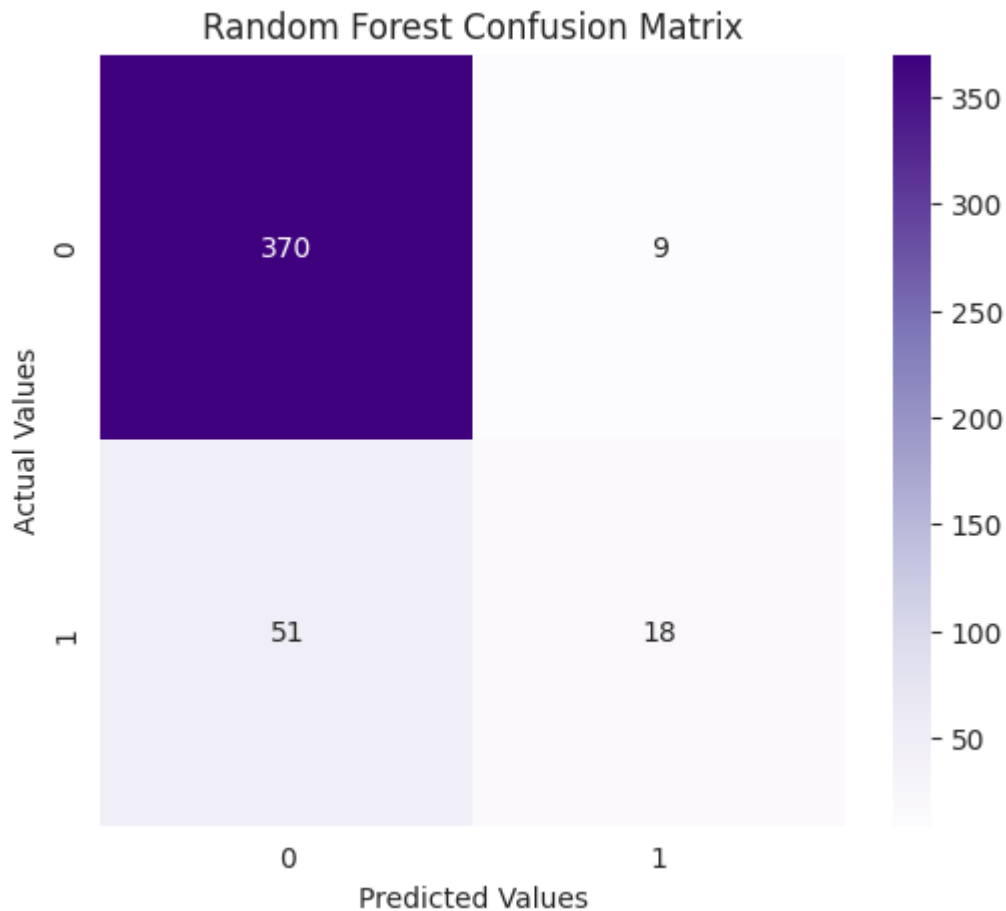


Figure 4.25: Random Forest Confusion Matrix

It is a measurement of accuracy of the Random Forest model, created by comparing the actual answers and how the model predicted them. The values that are obtained are as follows 370 true negatives, 18 true positives, 51 false negatives and 9 false positives. The model is very good in predicting the non-responder patients but not all positive responders. The customer groups vary primarily in regard to campaign engagement behaviour. In the case of automated decision systems, this implies that the system might require some extra methods to enhance the fairness and awareness of possible responders.

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.875000	0.685714	0.347826	0.461538
1	Decision Tree	0.796875	0.358974	0.405797	0.380952
2	Random Forest	0.866071	0.666667	0.260870	0.375000

Figure 4.26: Comparative Evaluation of Machine Learning Models

This measure is required to measure and compare the predictive accuracy, precision, recall, and F1 score for all the three models: Logistic regression, Decision tree and Random forest. The results obtained indicate that Logistic Regression has the highest accuracy (0.875) and F1-score (0.462) whereas, Decision Tree and Random Forest have lower values. The difference among customer groups is not directly determined as fairness of the group protection. For automated decision making, performance and fair treatment of the customer segments; have to be considered.

4.7 Feature Importance Analysis

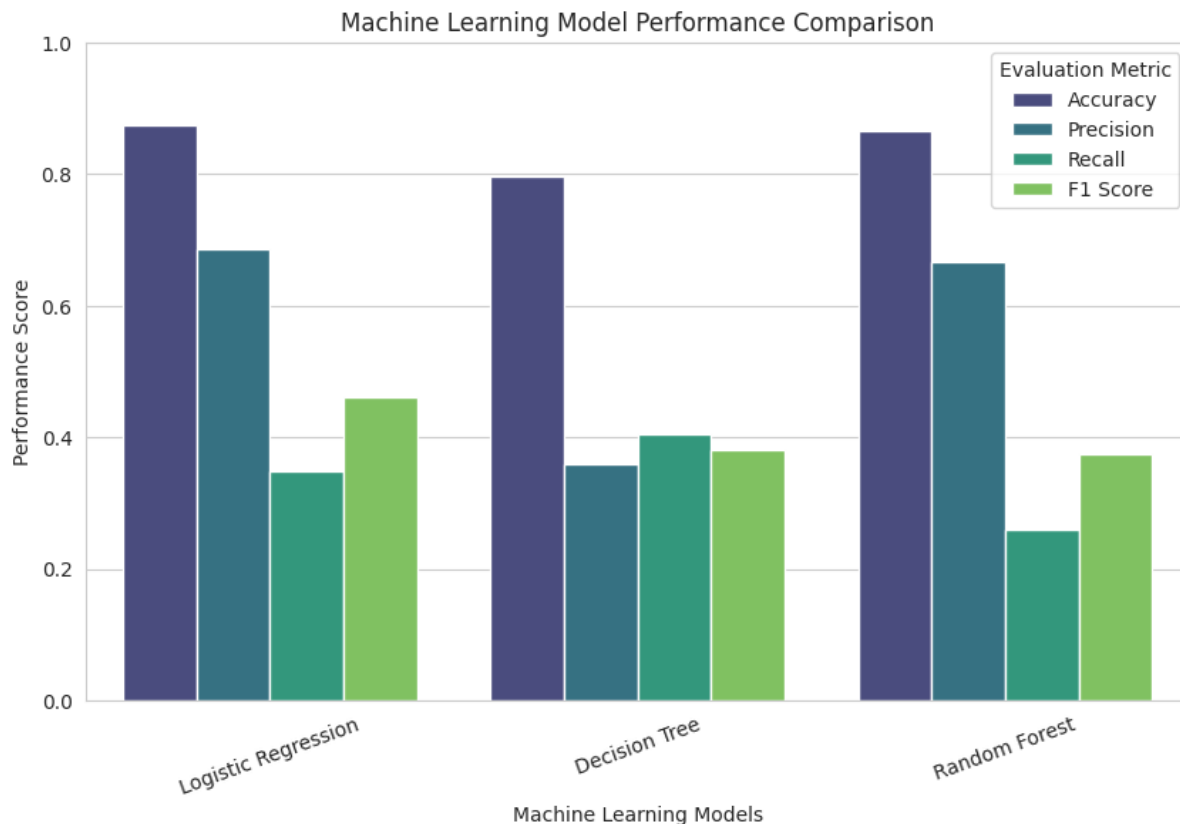


Figure 4.28: Machine Learning Model Performance Comparison

The figure is a comparison of Logistic Regression, Decision tree, and Random Forest model on basis of accuracy, precision, recall and F1-score. The highest accuracy (0.875) and F1-score (0.462) is obtained with the help of Logistic Regression although the same result is noted with Random Forest. Evaluation of the results of Decision Tree is rather poor.

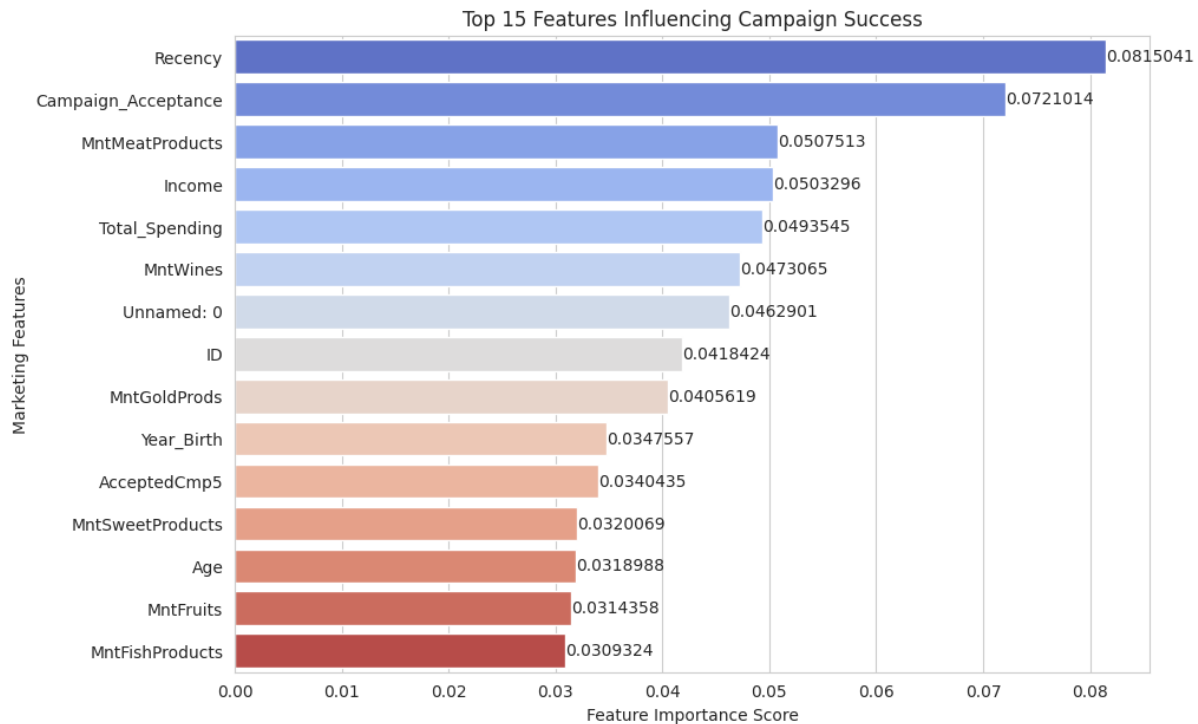


Figure 4.29: Top 15 Features Influencing Campaign Success (Feature Importance Analysis)

The figure is an evaluation of the relative significance of the input features of a computer-based model of the digital marketing campaign and its success. The resulting values of importance of features in the data means that the most important feature is Recency (0.0815), followed by Campaign_Acceptance (0.0721) MntMeatProducts (0.0508) Income (0.0503) and Total spendings (0.0494). These values serve to show the strength with which each feature is a predictor of results. The group of consumers differ in accordance with elements of their buying habits, cash flow and background of protected traits versus shielded personality characteristics.

4.8 Findings and Discussion

In this part it is not difficult to start talking about the findings from the quantitative analysis and how successful it is to achieve the research objective. The study aimed to investigate how predictive analytics and machine learning techniques can be applied to predict digital marketing campaign success. The secondary marketing campaign data set with 30 original attributes and 2240 customer records is analysed. The research has presented the important factors which impact the success of campaigns and has analysed the effectiveness of the models used, all in data preprocessing, feature engineering, exploration data analysis and predictive modelling using machine learning.

The first research question mitigated by that has been accomplished through exploratory data analysis and the feature importance analysis which found the key variables correlated with customers' response. The results of the feature importance indicated that Recency is the most dominant one having an importance score of 0.0815, the next one is Campaign_ Acceptance (0.0721), MntMeatProducts (0.0508), Income (0.0503), and Total spending (0.0494). The findings reveal that both customer engagement history and financial parameters are influential in predicting the success of campaigns and indicate that financial and buying behaviour are almost as significant as engagement history. Another finding in this analysis is that the customers who had been spending in a higher behaviour are more responsive to the campaign, this suggests that customers' spend history, in terms of engagement, is also a significant influencing factor in the effectiveness of digital marketing.

The second objective is to do data preprocessing and analytical preparation based on second marketing data. The dissertation completed the stage of data cleaning and dealing with missing values, and applying feature engineering, encoding and feature scale. It's a 2240 row by 30 column dataset in which the only data that is missing are 24 values in the Income column. Before development of the models, the missing values are detected and handled, which guaranteed a better quality of data. Other characteristics like customer age, total spending, total purchase and past campaign acceptance are added, to enhance predictive power. The final feature matrix consisted of 2,240 records and 35 predictive variables, which means that the machine learning algorithms are able to analyse customer behaviour more efficiently.

The third is to create machine learning models which can predict the success of digital marketing campaigns. Three of them – decision tree, random forest and logistic regression – are built in the form of classifiers. The quantitative analysis revealed that the models had differences in predictive performance. The accuracy obtained by Logistic Regression is the highest of 87.5 then the precision followed with 68.6 then recall with 34.8 followed by F1-score of 46.2. Random Forest achieved an accuracy of 86.6%, precision of 66.7%, recall of 26.1%, and F1-score of 37.5%. Decision Tree achieved an accuracy of 79.7%, precision of 35.9%, recall of 40.6%, and F1-score of 38.1%. This shows that the best overall predictive performance of this dataset is given by Logistic Regression.

The fourth aim is to assess the model performance based on proper classification measures. Accuracy, precision, recall and F1-score as well as confusion matrix analysis are used to accomplish this goal. The confusion matrix outputs indicated that the Logistic Regression is able to accurately classify 368 and 24 non responsive customers and responsive customers, respectively and yielded 45 false rejection and 11 false acceptance. By performing random forest classification, 370 non responsive customers and 18 responsive ones are classified correctly while a decision tree did the same for 329 non responsive customers and 28 responsive ones are correctly classified. The results of the studies show that despite the effectiveness of all models in identifying non-responsive customers, it is more difficult to predict positive campaign outcomes as there are three times more negative outcomes of the campaign.

In general, the numerical evidence confirms the fact that predictive analytics and machine learning could offer valuable information to optimise digital marketing campaigns. The following are the aspects that are accomplished in the research objectives: 1) Identifying / determining the factors of importance (factor oriented), 2) Enhancing the quality of the data by preprocessing, 3) Building forecasting models, and 4) Making comparisons with the performance of the built forecasting models. The results confirm that information on previous contact with the company and on their purchasing behaviour and expenditure, are valuable indicators of whether a campaign will work. However, organisations must use caution when taking advice from models that predict the trends based on the dataset, as there may be relations with the data with factors that are not demographic, such as age, income and education. The findings have shown that data-based strategies can be used

to improve marketing decisions, marketing resources allocation, and marketing campaign planning.

4.9 Research Novelty

This research novelty utilizes a model that is based on combined predictive analytics in relation to machine learning algorithms and implementation of a digital marketing campaign to see its effectiveness. This study combines the four aspects of data pre-processing, feature engineering, exploratory analysis, machine learning modelling and performance measurement in the same platform, while previous studies have mainly concentrated on exploring the behaviour of customers through segmentation or specific prediction methods.

One of the main contributions of this research is the comparative analysis of three machine learning models (Logistic Regression, Decision Tree, and Random Forest) based on several performance metrics such as accuracy, precision, recall, F1-score and the analysis of a confusion matrix. This gives a more comprehensive picture of the effectiveness of models to predict the results of campaign as opposed to excluding a single predictive strategy.

The other is the application of feature engineering techniques to transform the raw customer data into the new meaningful predictive variables like total spending, total purchases, customer age and campaign acceptance indicators, etc. This improves the capacity that machine learning models have to learn the behaviour patterns of customers. It generally provides a contribution by relating predictive accuracy and insights from the study that any organisation could use to make more informed decisions when optimising their campaigns, market to their customers, and allocate resources.

4.10 Chapter Summary

This chapter presented the results of the quantitative research that is undertaken to make a prediction of the success of the digital marketing campaign. In all cases, descriptive analysis, preprocessing, exploratory data analysis and feature engineering are used to find out and to refine the data for enhanced predictive accuracy. The analysis resulted in identification of some of the important variables that would determine the success of the campaigns namely recency of contact, acceptance of campaigns, spending patterns of the customers, their income and purchasing

patterns. Evaluation metrics are used to develop and compare three models of machine learning, which are: Logistic Regression, Decision Tree and Random Forest. The highest overall accuracy and F1-score of all the considered models is obtained with Logistic Regression. The significance of the features analysis also helped to identify the variables used for prediction of a campaign. The results support the effectiveness of predictive analytics in helping to optimise marketing and make decision-making based on data. The study is summarized in the next chapter and recommendations derived from the study are provided in the final chapter.

Chapter 5: Conclusion and Recommendations

5.1 Conclusion

This research paper discusses how the predictive analytics and machine learning techniques can be utilized to predict success with a digital marketing campaign. The research focused on knowing key marketing performance variables, forming predictive models, and quantifying their usefulness in aiding evidence-based marketing choices. Data for the study comes from a secondary marketing data set, data preprocessing, exploratory data analysis, feature engineering, and machine learning techniques, to get customer behaviour and campaign results.

The results showed the possibilities of predictive analytics that could be useful in the evaluation of the performance of digital marketing, drawing patterns in customer data. The analysis of feature importance demonstrated that, the aspects like recency, past campaign acceptance, customer spending behaviour, income, and total spending had significant impacts on campaign response prediction. Three machine learning models (Logistic Regression, Decision Tree, and Random Forest) are created and compared. Out of these models, Logistic Regression is observed to be able to achieve the best overall performance with 87.5% accuracy and 46.2 F1 score and is observed to be effective on this data.

The study highlights that organisations should bid farewell to the conventional study methodology and look into predictive analytics to foresee possible campaign stretches and the utilization of resources. However, the results indicate that there are potentially problematic issues that arise from an undervalue estimation of "positive" campaign reaction. Overall, it shows that digital marketing optimization and improved customer targeting strategies can be supported using machine learning based predictive analytics and that organisations can get evidence-based information, which will enable them to make informed decisions.

5.2 Linking with the Objectives

The first objective of the research is to identify important digital marketing factors that make a digital marketing campaign successful. This has been set by applying exploratory data analysis and using the importance of features. The findings result in the following: Recency, campaign

acceptance, customer spending behavior, income and purchasing behavior are found to be important at predicting campaign results.

The second objective is to perform Pre-processing and analytical preparations of the data using secondary market data. The data cleaning, missing values, encoding, feature engineering, and feature scaling helped in accomplishing this goal. The conversion of raw customer data into meaningful variables enhanced the usefulness and quality of the data in analysing through machine learning.

The third aim is to devise machine learning to lead the success of the campaign. Logistic Regression, Decision trees and Random Forest are successfully developed and tested models. The comparison showed that there are differences in the predictive ability, whereby the Logistic Regression has the highest overall performance.

The fourth willingness is that of finding out the performance of the models using suitable classification metrics. This has been done via accuracy, precision, recall, F1-score and confusion matrix. The results provided a quantitative comparison of the models and showed the strengths and weaknesses of each of the models.

Overall, the consolidation of all the aims of the research in predictive modelling approaches and marketing analysis is achieved. The research is able to show how machine learning methods can enhance the knowledge of campaign success, and promote data-driven digital marketing.

5.3 Recommendations

Recommendation 1: Improve Data Quality and Feature Development

The best advice that can be offered to an organisation would be to collect good quality and abundance of marketing data to improve the accuracy of prediction. In the future the models for predicting behavior should be expanded so that other behavioral traits, like customer interaction history, interaction with the web, engagement pattern, and frequency of purchases are taken into account (GhorbanTanhaei *et al.*, 2024). Developing features better can enhance machine learning algorithms to recognize the potential customers and more accurately predict campaign responses.

Recommendation 2: Apply Advanced Machine Learning Techniques

The future marketing analytics systems are advised to delve into more sophisticated machine learning methods including using gradient boosting, neural networks and ensemble computing.

Logistic Regression performed fairly well in this analysis but there may be more complex algorithms that can understand the complex relationship in the customer behaviour data points (Sunarya *et al.*, 2024). Various models should be assessed by organisations periodically, to identify appropriate methods to use on particular marketing goals.

Recommendation 3: Use Predictive Insights for Personalised Marketing Strategies

Predictive analytics results should be used by businesses to create personalised marketing campaigns. Predictive models can be used to identify the customer behaviour patterns that make it able to facilitate targeted advertising, enhance customer engagement, and effective allocation of marketing resources (GhorbanTanhaei *et al.*, 2024). However, segmenting the customer the decision-making has to be understandable and in organisations an economical proportion of requirement to demographic correlating and dependency should be minimised.

5.4 Theoretical and Practical Implications

Theoretical Implications

The study will contribute to the available details of predictive analytics and digital marketing, as it will show how machine learning methods can be utilized to predict the effects of a campaign. In the study, it builds on the past findings having combined predictive modelling, analyses of features importance and comparative analysis to comprehend the factors that affect marketing performance.

Practical Implications

The practical implication of this study is that organisations can use the systematic approach to the implementation of predictive analytics in digital marketing in practice. The findings of the study have revealed that attributes of customers, such as their extent of recency, ways they spend money and whether they have previously responded to a campaign can be used more effectively for targeting and planning a campaign. The findings can help marketing managers to effectively allocate resources, enhance customer engagement and make evidence-based decisions. Organisations are however advised to constantly observe model performance, and train predictive systems as the behaviour of customers evolves with time.

5.5 Research Contribution

The study will contribute to the digital marketing analytics field by providing a consistent framework that spans from the preprocessing of data, incorporation of feature engineering, exploratory analysis, and use of machine learning to perform the prediction process. Using Logistic Regression, Decision Tree and Random Forest models, this paper compares the three models based on their abilities to predict the success of the camping campaign. Moreover, the importance of the features analysis will determine the most important variables of customer reactions and give predictive and management information. This is contrary to other studies that only address accuracy of the prediction, but the outcomes of the models are not related to the actual marketing decision making. The results add to the knowledge of how predictive analytics could be used to optimise the campaigns, target customers, and allocate the resources better.

5.6 Research Limitation

This study has a number of shortcomings. The research is a secondary one, so there is no control for the quality of data collection and variables involved in this research. The data set might not reflect digital marketing settings and target markets. In addition, there is the testing of only three machine learning models and the real-time campaign data is not considered. The limitations of this research need to be overcome with bigger and more heterogeneous datasets in the future.

5.7 Future Scope

There are several ways this study could be enhanced for future use, such as implementing state of the art machine learning and deep learning models with utilizing large real-time marketing datasets. There are many other behavioural variables, interactions on social media and information about a customer's journey that can be added in the future research so the prediction is more accurate. The automated recommendation systems which may give personalised marketing strategies based on predictive customer knowledge can also be researched.

References

- Al Khaldy, M.A., Al-Obaydi, B.A.A. and al Shari, A.J., (2023, May). The impact of predictive analytics and AI on digital marketing strategy and ROI. In *Conference on sustainability and cutting-edge business technologies* (pp. 367-379). Cham: Springer Nature Switzerland.
- Al Rafi, M., (2023). Machine learning-enhanced predictive marketing analytics for optimizing customer engagement and sales forecasting. *International Journal of Research and Applied Innovations*, 6(4), pp.9203-9213.
- Alam, S., Ayub, M.S., Arora, S. and Khan, M.A., (2023). An investigation of the imputation techniques for missing values in ordinal data enhancing clustering and classification analysis validity. *Decision Analytics Journal*, 9, p.100341.
- Almestarihi, R., Ahmad, A.Y., Frangieh, R.H., Abualsondos, I.A., Nser, K.K. and Ziani, A., (2024). Measuring the ROI of paid advertising campaigns in digital marketing and its effect on business profitability.
- Bellar, O., Baina, A. and Ballafkih, M., (2024). Sentiment analysis: Predicting product reviews for e-commerce recommendations using deep learning and transformers. *Mathematics*, 12(15), p.2403.
- Berlilana, B., Hariguna, T. and El Emary, I.M., (2025). Enhancing digital marketing strategies with machine learning for analyzing key drivers of online advertising performance. *Journal of Applied Data Sciences*, 6(2), pp.817-827.
- Bohara, S.R., (2025). Statistics as a tool for market research and consumer behavior analysis. *KVM Research Journal*, 6, pp.147-156.
- Chattopadhyaya, A., (2026). Financial Analytics in Digital Marketing Campaign Optimization. *Scriptora International Journal of Research and Innovation (SIJRI)*, pp.39-46.

Cruz, R.N. and Rosário, A.T., (2025). Data-driven decision-making in marketing: A systematic literature review of emerging themes and research gaps. *Systems*, 13(12), p.1114.

Cynthia, M.M., Syahri, R., Rangkuti, M.H.A.G., Firdaus, M.A., Waruwu, R.F.S., Halawa, Y. and Ritonga, T., (2025). Prediction of Digital Marketing Campaign Success Using Deep NeuralNetwork Models. *Jurnal Sistem Informasi dan Teknologi Jaringan*, 6(2), pp.49-53.

Dutta, D.C., (2024). A Study On The Factors That Influence The Success Of Digital Marketing In A Dynamic Marketing Landscape. *A Theoretical And Empirical Integration, Educational Administration: Theory And Practice*, 30(4), pp.3793-3798.

Elgendy, N., Elragal, A. and Päivärinta, T., (2022). DECAS: a modern data-driven decision theory for big data and analytics. *Journal of Decision Systems*, 31(4), pp.337-373.

El-Hajj, M. and Pavlova, M., (2024). Predictive modeling of customer response to marketing campaigns. *Electronics*, 13(19), p.3953.

El-Hajj, M. and Pavlova, M., (2024). Predictive modeling of customer response to marketing campaigns. *Electronics*, 13(19), p.3953.

El-Hajj, M. and Pavlova, M., (2024). Predictive modeling of customer response to marketing campaigns. *Electronics*, 13(19), p.3953.

El-Hajj, M. and Pavlova, M., (2024). Predictive modeling of customer response to marketing campaigns. *Electronics*, 13(19), p.3953.

Faraz, N. and Anjum, A., (2025). Spendception: The psychological impact of digital payments on consumer purchase behavior and impulse buying. *Behavioral Sciences*, 15(3), p.387.

García-Salirrosas, E.E. and Rondon-Eusebio, R.F., (2022). Green marketing practices related to key variables of consumer purchasing behavior. *Sustainability*, 14(14), p.8499.

Ghodake, S.P., Malkar, V.R., Santosh, K., Jabasheela, L., Abdufattokhov, S. and Gopi, A., (2024). Enhancing Supply Chain Management Efficiency: A Data-Driven Approach using Predictive

Analytics and Machine Learning Algorithms. *International Journal of Advanced Computer Science & Applications*, 15(4), p.672.

GhorbanTanhaei, H., Boozary, P., Sheykhan, S., Rabiee, M., Rahmani, F. and Hosseini, I., (2024). Predictive analytics in customer behavior: Anticipating trends and preferences. *Results in Control and Optimization*, 17, p.100462.

GhorbanTanhaei, H., Boozary, P., Sheykhan, S., Rabiee, M., Rahmani, F. and Hosseini, I., (2024). Predictive analytics in customer behavior: Anticipating trends and preferences. *Results in Control and Optimization*, 17, p.100462.

Gülter, E. and Cevher, M.F., (2025). Evolution of digital marketing campaigns with artificial intelligence and machine learning: Analysing success prediction capabilities. *Business & Management Studies: An International Journal*, 13(2), pp.478-493.

Gupta, S., Kishan, B. and Gulia, P., (2024). Comparative analysis of predictive algorithms for performance measurement. *IEEE Access*, 12, pp.33949-33958.

Gupta, S., Kishan, B. and Gulia, P., (2024). Comparative analysis of predictive algorithms for performance measurement. *IEEE Access*, 12, pp.33949-33958.

Haque, B.T. and Rahman, M.A., (2024). Data Preparation and Feature Engineering in High-Dimensional Predictive Modeling. *Journal of Sustainable Development and Policy*, 3(04), pp.205-244.

Hariguna, T. and Chen, S.C., (2024). Customer segmentation and targeted retail pricing in digital advertising using gaussian mixture models for maximizing gross income. *Journal of Digital Market and Digital Currency*, 1(2), pp.183-203.

Hayadi, B.H. and El Emary, I.M., (2024). Predicting campaign ROI using decision trees and random forests in digital marketing. *Journal of Digital Market and Digital Currency*, 1(1), pp.1-20.

Ikeh, C. O. (2025). Machine learning in digital marketing: Real-time campaign optimization and conversion prediction using multimodal consumer interaction data. *International Journal of Computer Applications Technology and Research*, 14(05), 39-54.

Inglis, A., Parnell, A. and Hurley, C.B., (2022). Visualizing variable importance and variable interaction effects in machine learning models. *Journal of Computational and Graphical Statistics*, 31(3), pp.766-778.

Ittransition (2026), Predictive analytics in marketing: a technology & market state overview. Available at: <https://www.ittransition.com/predictive-analytics/marketing#:~:text=Optimize%20your%20marketing%20efforts%20with> (Accessed: 18 August 2026)

Jalal, N., Mehmood, A., Choi, G.S. and Ashraf, I., (2022). A novel improved random forest for text classification using feature ranking and optimal number of trees. *Journal of King Saud University-Computer and Information Sciences*, 34(6), pp.2733-2742.

Kaponis, A., Maragoudakis, M. and Sofianos, K.C., (2025). Enhancing user experiences in digital marketing through machine learning: Cases, trends, and challenges. *Computers*, 14(6), p.211.

Kim, A., Öström, Å., Mihnea, M. and Niimi, J., (2024). Consumers' attachment to meat: Association between sensory properties and preferences for plant-based meat alternatives. *Food Quality and Preference*, 116, p.105134.

Malik, S. and Rana, A., (2025). Transforming Digital Marketing with Machine Learning Algorithms. *Journal of Graphic Era University*, pp.271-300.

Marković, A., (2026). THE APPLICATION OF DATA ANALYTICS IN THE OPTIMISATION OF DIGITAL MARKETING CAMPAIGNS. *SCIENCE International Journal*, 5(2), pp.245-250.

Massoudi, A.H., Birdawod, H.Q. and Raewf, M.B., (2023). Personal digital marketing influence on successful marketing campaign in today's digital age. *Cihan University-Erbil Journal of Humanities and Social Sciences*, 7(1), pp.158-165.

Miah, M.R. and Islam, M.M., (2024). Data Preprocessing and Feature Engineering Strategies for Large-Scale Predictive Modeling Applications. *Review of Applied Science and Technology*, 3(01), pp.263-302.

Miah, M.R. and Islam, M.M., (2024). Data Preprocessing and Feature Engineering Strategies for Large-Scale Predictive Modeling Applications. *Review of Applied Science and Technology*, 3(01), pp.263-302.

Modak, C., Ghosh, S.K., Sarkar, M.A.I., Sharif, M.K., Arif, M., Bhuiyan, M., Ahmed, M.P., Pabel, M.A.H. and Devi, S., (2024). Machine learning model in digital marketing strategies for customer behavior: harnessing CNNs for enhanced customer satisfaction and strategic decision-making. *Journal of Economics, Finance and Accounting Studies*, 6(3), pp.178-186.

Mokari, A., Guo, S. and Bocklitz, T., (2023). Exploring the steps of infrared (IR) spectral analysis: Pre-processing,(classical) data modelling, and deep learning. *Molecules*, 28(19), p.6886.

Mukred, M., Asma'Mokhtar, U., Hawash, B., AlSalman, H. and Zohaib, M., (2024). The adoption and use of learning analytics tools to improve decision making in higher learning institutions: An extension of technology acceptance model. *Heliyon*, 10(4).

Mundu, M.M., Sempewo, J.I., Goparaju, A. and Uti, D.E., (2026). Comparative analysis of model evaluation metrics in energy systems, environmental modeling, and sustainability science. *International Journal of Energy Research*, 2026(1), p.6170467.

Muraina, I.O., Adesanya, O.M., Agoi, M.A. and Onen, S., (2023). The Necessity of Exploratory Data Analysis: How are preprocessing activities beneficial to Data Analysts and Professional Researchers in Academia?. *Int. J. Sci. Res. in Computer Science and Engineering*, 11(3).

Omar, E. D., Mat, H., Abd Karim, A. Z., Sanaudi, R., Ibrahim, F. H., Omar, M. A., ... and Goh, B. L. (2024). Comparative analysis of logistic regression, gradient boosted trees, SVM, and random forest algorithms for prediction of acute kidney injury requiring dialysis after cardiac surgery. *International journal of nephrology and renovascular disease*, 197-204.

Osakwe, J., Shilongo, A. and Ziezo, M., (2023, December). Optimising customer segmentation in digital marketing using predictive analytics: A review of literature. In *Proceedings of the International Conference on Information Systems and Emerging Technologies (ICISSET)*.

Paşcalău, S.V., Popescu, F.A., Bîrlădeanu, G.L. and Gigauri, I., (2024). The effects of a digital marketing orientation on business performance. *Sustainability*, 16(15), p.6685.

Phuong Dung, P.T., Minh An, H., Huy, P.Q. and Dinh Quy, N.L., (2023). Understanding the startup's intention of digital marketing's learners: An application of the theory of planned behavior (TPB) and technology acceptance method (TAM). *Cogent Business & Management*, 10(2), p.2219415.

Prakash, D., (2024). Data-driven management: The impact of big data analytics on organizational performance. *International Journal for Global Academic & Scientific Research*, 3(2), pp.12-23.

Prasetio, A.B., bin Mohd Aboobaider, B. and bin Ahmad, A., (2025). Predicting customer conversion in digital marketing: Analyzing the impact of engagement metrics using logistic regression, decision trees, and random forests. *Journal of Digital Market and Digital Currency*, 2(2), pp.181-204.

Rahardja, U. and Aini, Q., (2025). Evaluating the effectiveness of digital marketing campaigns through conversion rates and engagement levels using anova and chi-square tests. *Journal of Digital Market and Digital Currency*, 2(1), pp.26-45.

Rahman, M.A. and Haque, B.T., (2022). Quantitative Benchmarking of Machine Learning Models for Risk Prediction: A Comparative Study Using AUC/F1 Metrics and Robustness Testing. *Review of Applied Science and Technology*, 1(03), pp.32-60.

Ranjan, M., Barot, K., Khairnar, V., Rawal, V., Pimpalgaonkar, A., Saxena, S. and Sattar, A.M., (2023). Python: empowering data science applications and research. *Journal of Operating Systems Development & Trends*, 10(1), pp.27-33.

Roustaei, N., (2024). Application and interpretation of linear-regression analysis. *Medical Hypothesis, Discovery and Innovation in Ophthalmology*, 13(3), p.151.

Sadrnia, L. (2023). The Future of Marketing: How Predictive Modeling Optimizes Campaign Strategies. *iBusiness*, 15(4), 249-262.

Sangsawang, T. (2024). Predicting ad click-through rates in digital marketing with support vector machines. *Journal of Digital Market and Digital Currency*, 1(3), 225-246.

Sapkota, R., Meng, Z., Churuvija, M., Du, X., Ma, Z. and Karkee, M., (2026). Comprehensive performance evaluation of yolov12, yolo11, yolov10, yolov9 and yolov8 on detecting and counting fruitlet in complex orchard environments. *Agriculture Communications*, p.100125.

Soltaninejad, M., Aghazadeh, R., Shaghaghi, S. and Zarei, M., (2024). Using machine learning techniques to forecast Mehrnam company's sales: A case study. *Journal of Business and Management Studies*, 6(2), pp.42-53.

Song, Y., Qiu, X. and Liu, J., (2025). The impact of artificial intelligence adoption on organizational decision-making: An empirical study based on the technology acceptance model in business management. *Systems*, 13(8), p.683.

Sunarya, P.A., Rahardja, U., Chen, S.C., Lic, Y.M. and Hardini, M., (2024). Deciphering digital social dynamics: A comparative study of logistic regression and random forest in predicting e-commerce customer behavior. *Journal of Applied Data Sciences*, 5(1), pp.100-113.

Sylvestre, C.E., (2024). The role of data-driven decision making in business strategy. *Research Output Journal of Education*, 3(3), pp.80-84.

Theodorakopoulos, L. and Theodoropoulou, A., (2024). Leveraging big data analytics for understanding consumer behavior in digital marketing: A systematic review. *Human behavior and emerging technologies*, 2024(1), p.3641502.

Tousi, A. and Luján, M., (2022). Comparative analysis of machine learning models for performance prediction of the spec benchmarks. *IEEE Access*, 10, pp.11994-12011.

Wan, T.T. and Wan, H.S., (2023). Predictive analytics with a transdisciplinary framework in promoting patient-centric care of polychronic conditions: Trends, challenges, and Solutions. *AI*, 4(3), pp.482-490.

Wang, Y., Cao, M., Fan, Z. and Peng, S., (2022, June). Learning to detect 3D facial landmarks via heatmap regression with graph convolutional network. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 36, No. 3, pp. 2595-2603).

Wirtz, B.W. and Balzer, I., (2023). Social Media Marketing-A Systematic Review of Quantitative Empirical Studies. *Journal of Business and Management*, 29(1), pp.80-114.

Zhang, Y., Safdar, M., Xie, J., Li, J., Sage, M. and Zhao, Y.F., (2023). A systematic review on data of additive manufacturing for machine learning applications: the data quality, type, preprocessing, and management. *Journal of Intelligent Manufacturing*, 34(8), pp.3305-3340.

Ziakis, C. and Vlachopoulou, M., (2023). Artificial intelligence in digital marketing: Insights from a comprehensive review. *Information*, 14(12), p.664.

Appendices

Appendix 1: GitHub Link

<https://github.com/Anmollgupta/Thesis>